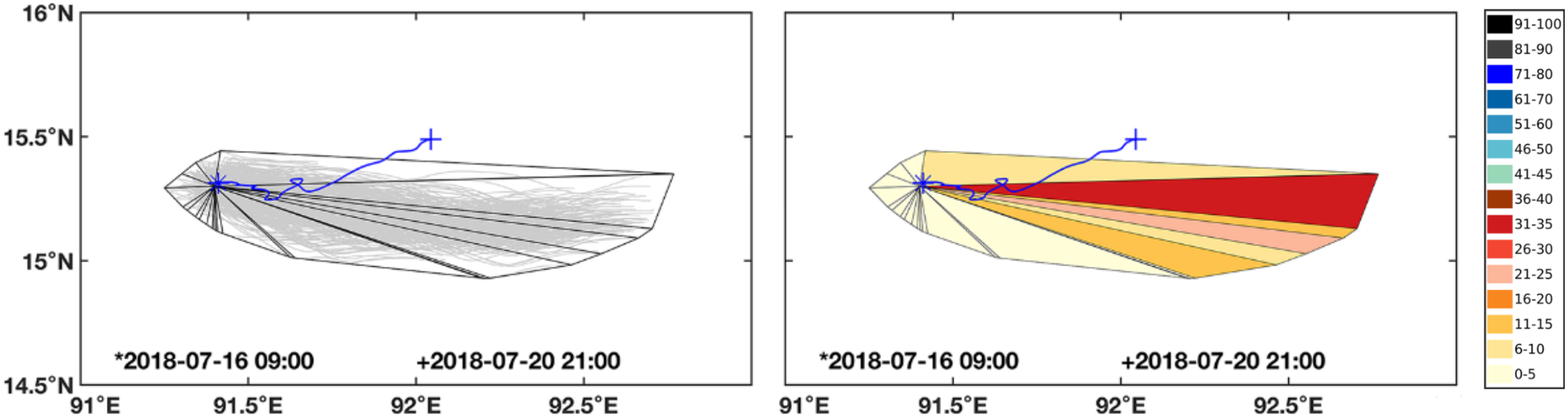
Search and Aid Rescue Tool (SARAT) version 3

Major model improvements and visualization additions

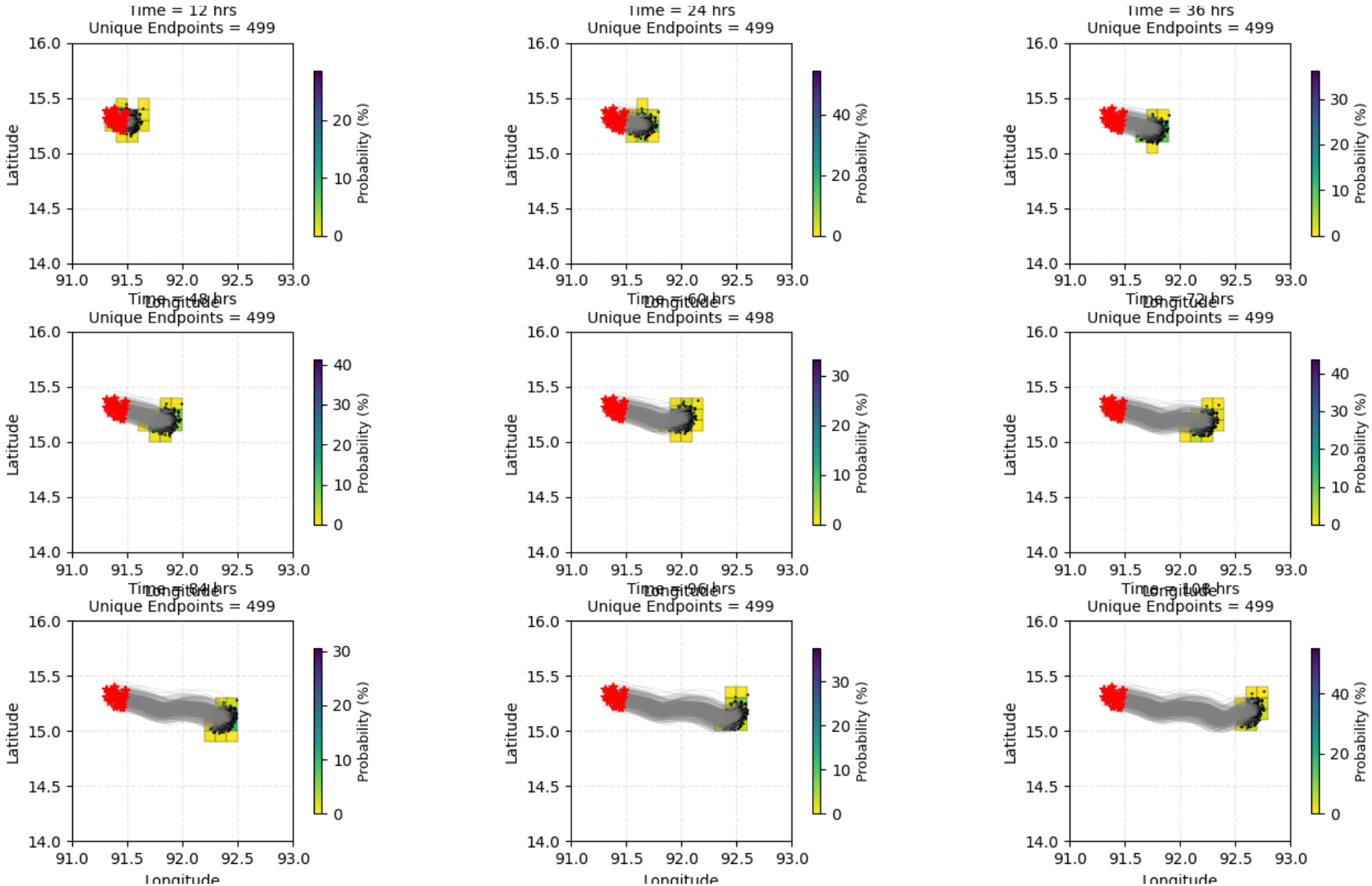
P Vijay, TNC Karthik, Arkaprava R, M Pravallika, and D Manasa

Simulated particle trajectories and probability calculation in SARAT version 2



- Particles are sprinkled throughout the simulation period: ideal for a constant release of objects
- Probabilities are calculated for the whole period: suitable for reaching a destination (e.g., beaching)
- Lacks the time evolution of probabilities

Simulated particle trajectories and probability calculation in SARAT version 3: Case 1



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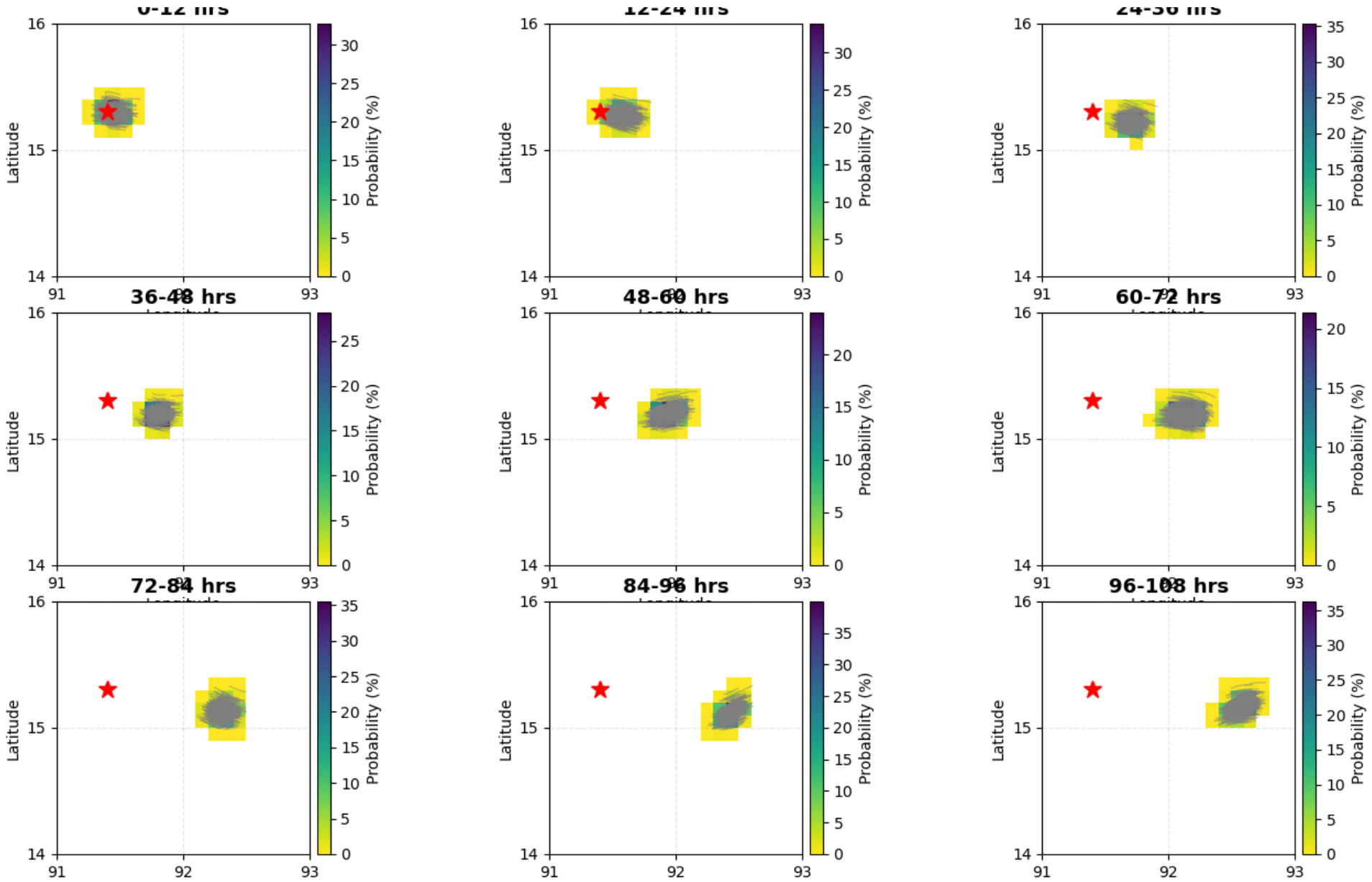

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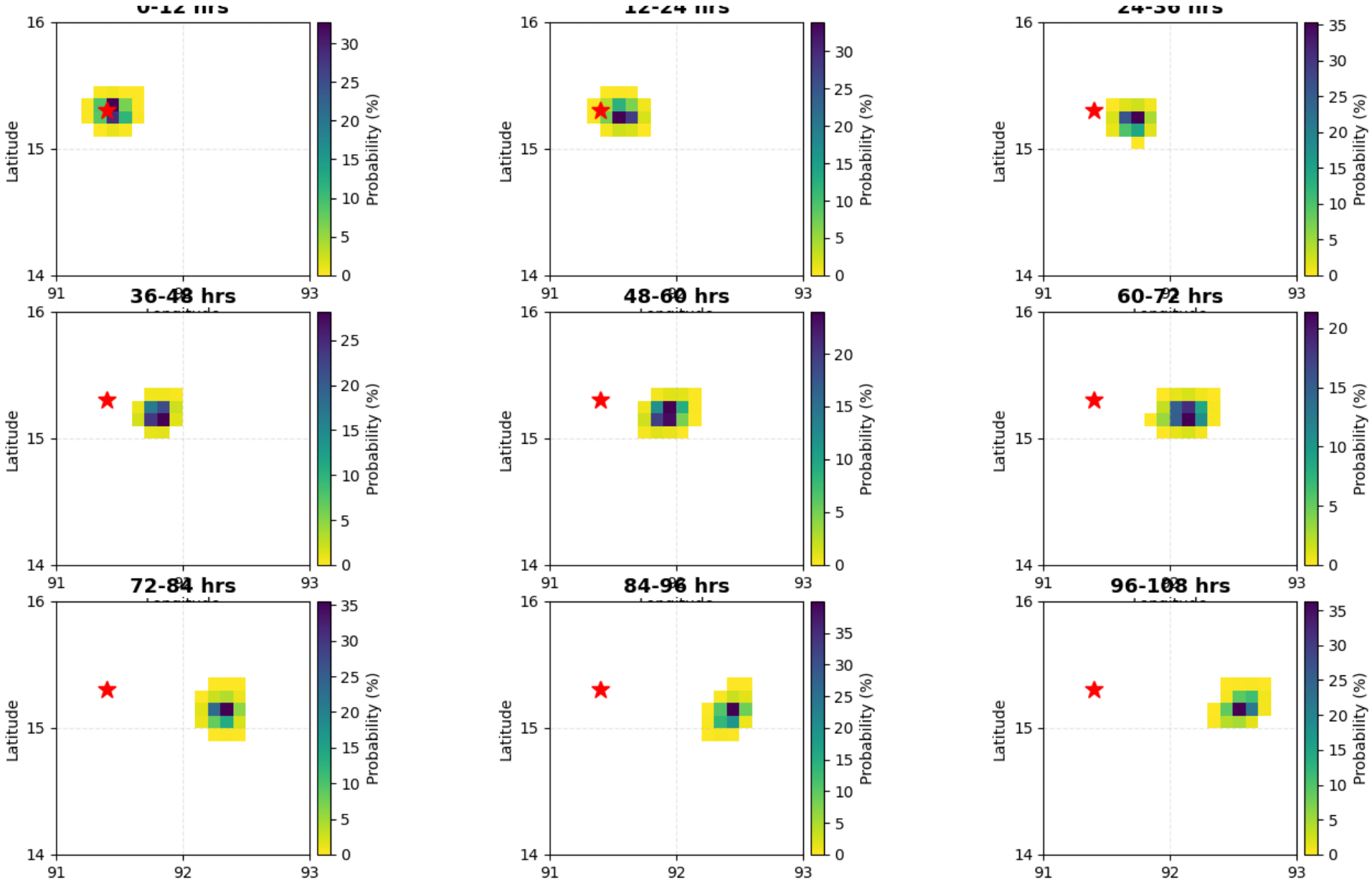
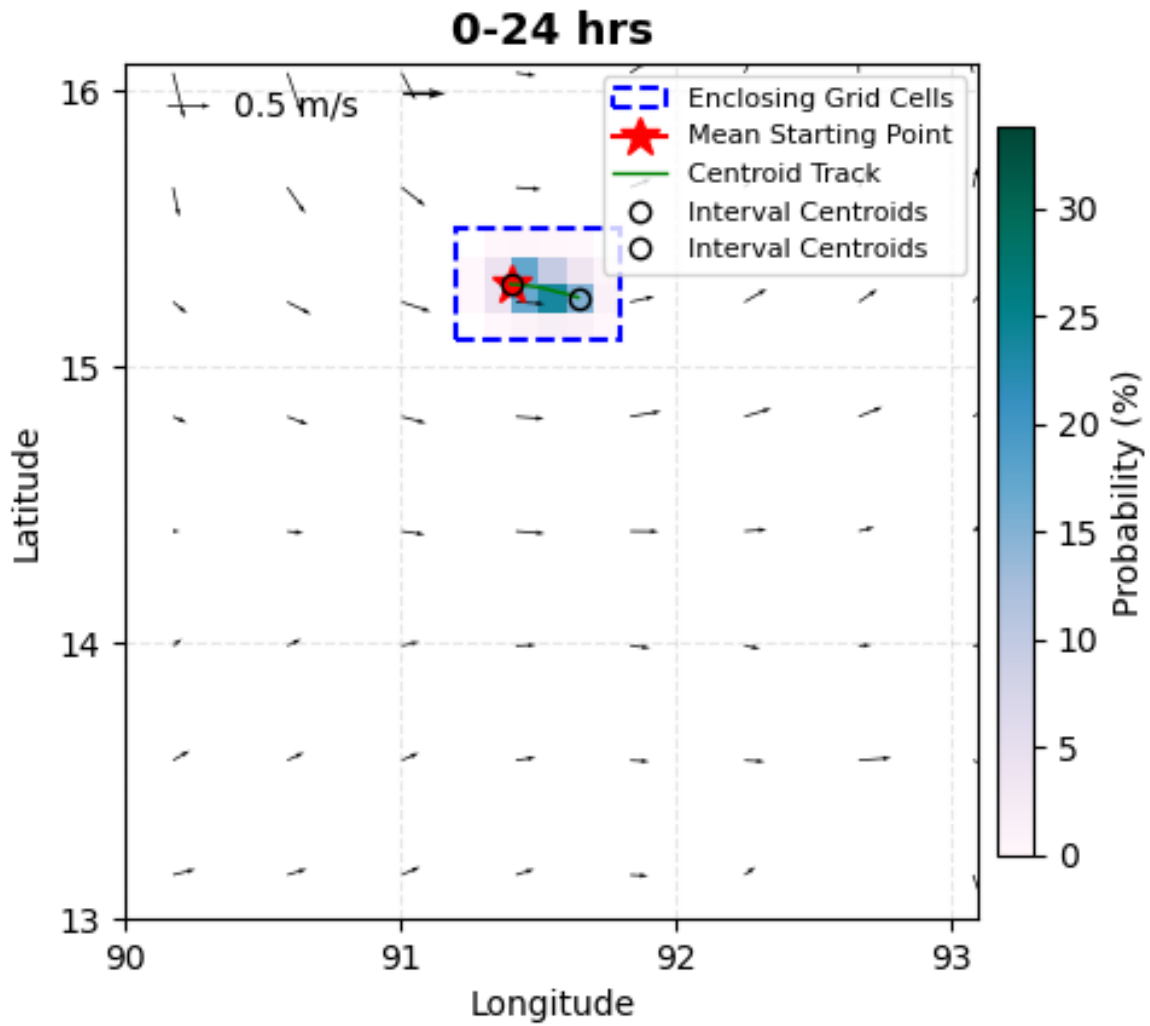


Time evolution of probability map in SARAT version 3: Case 1



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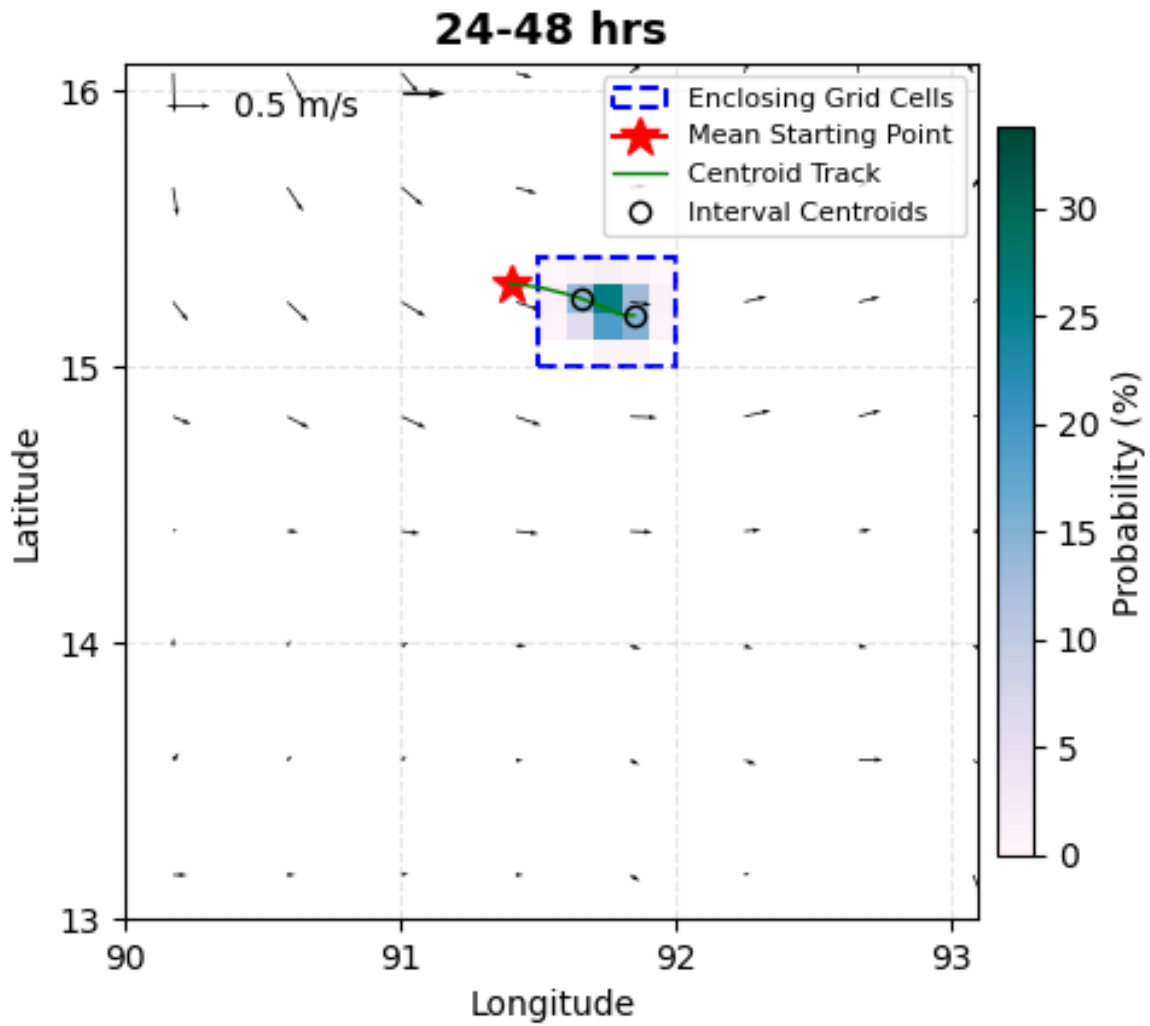

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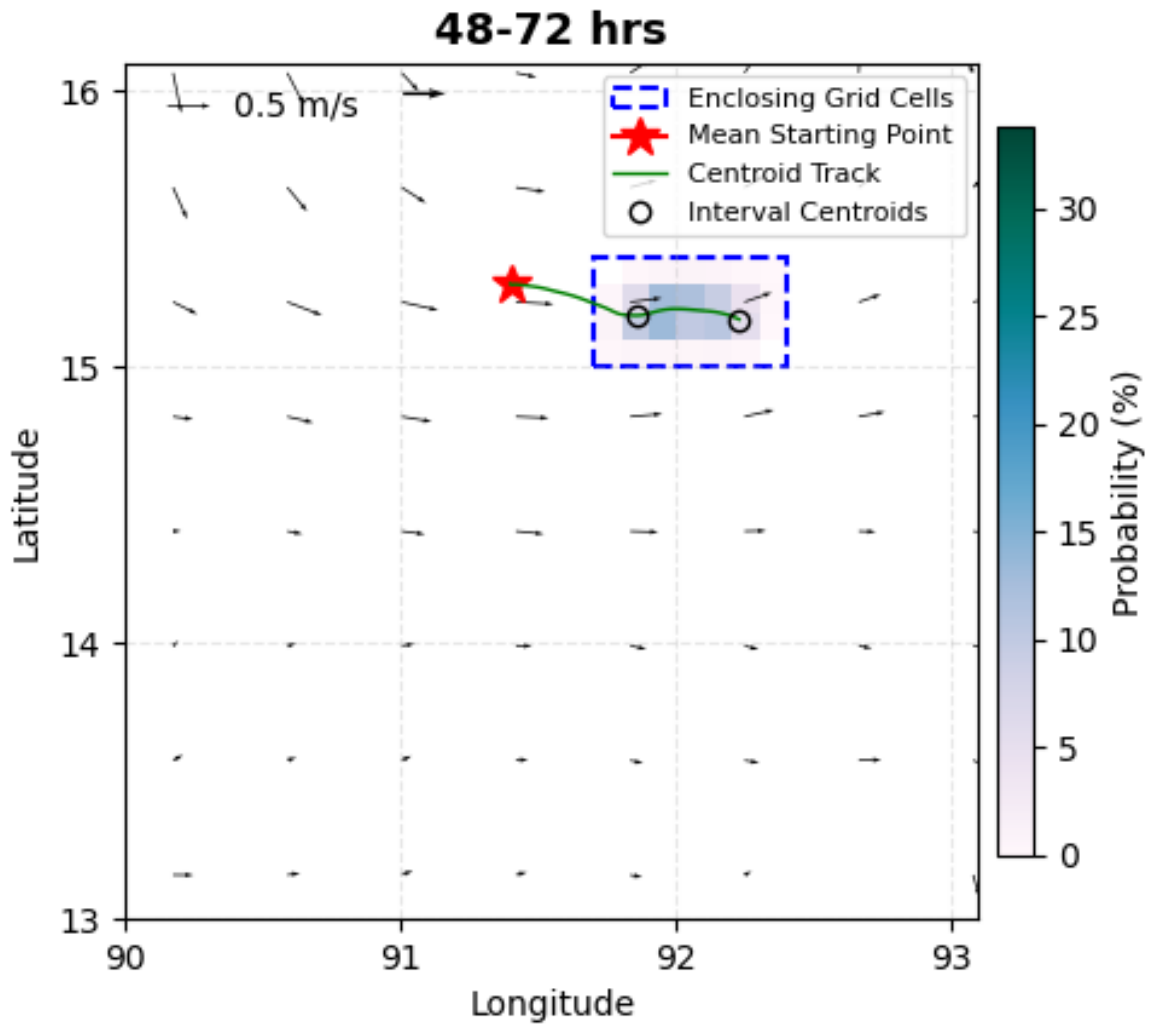

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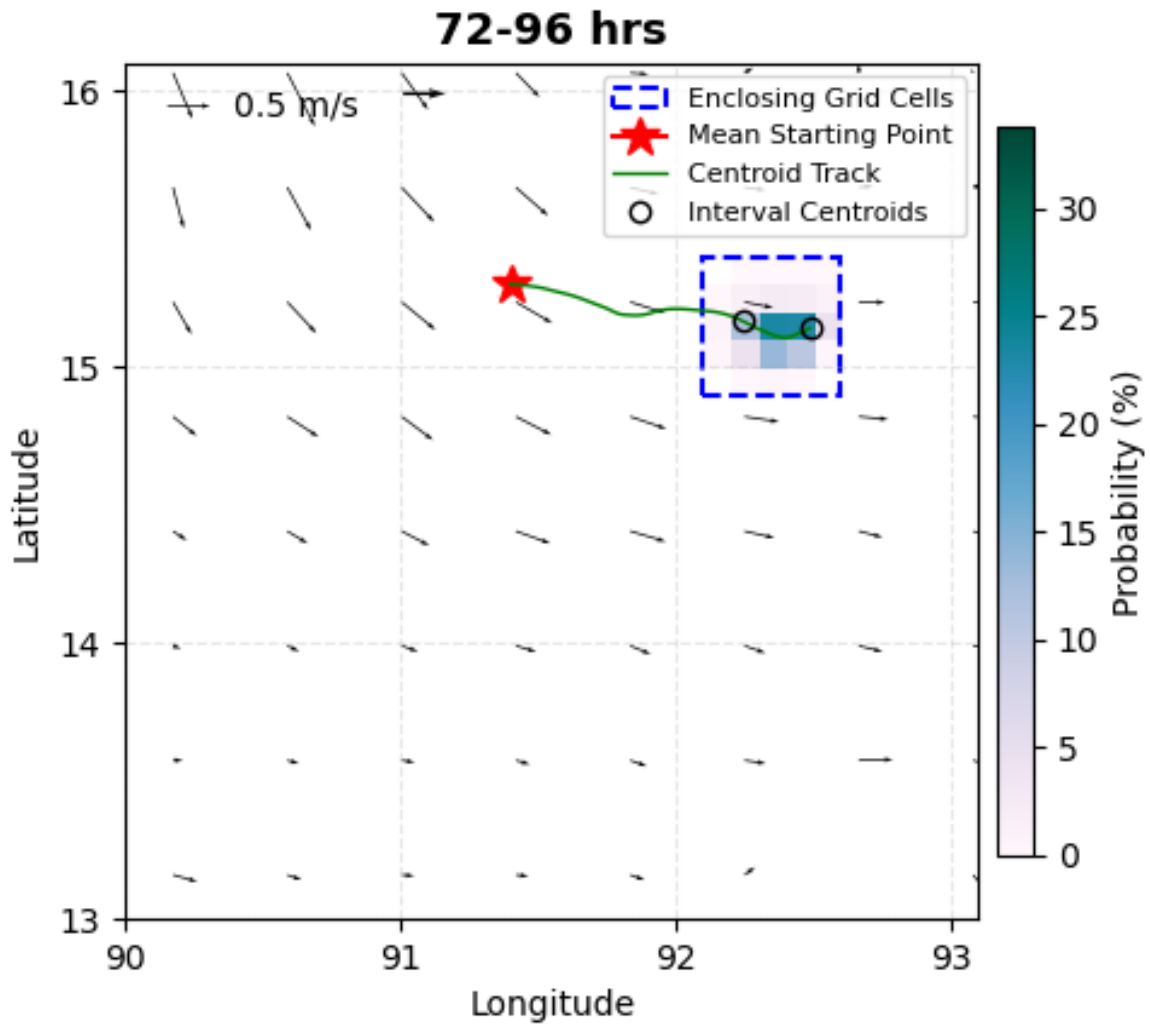

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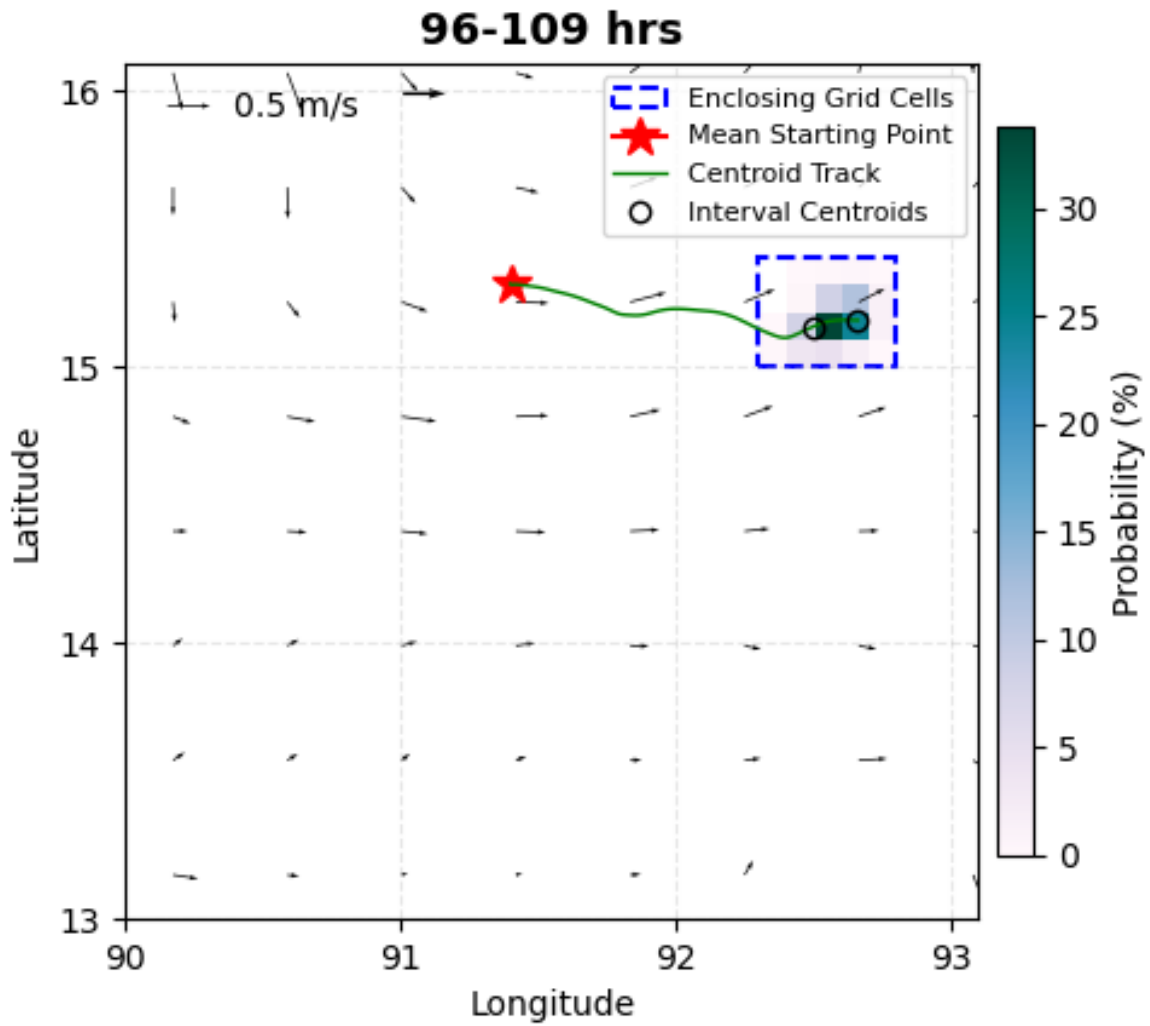
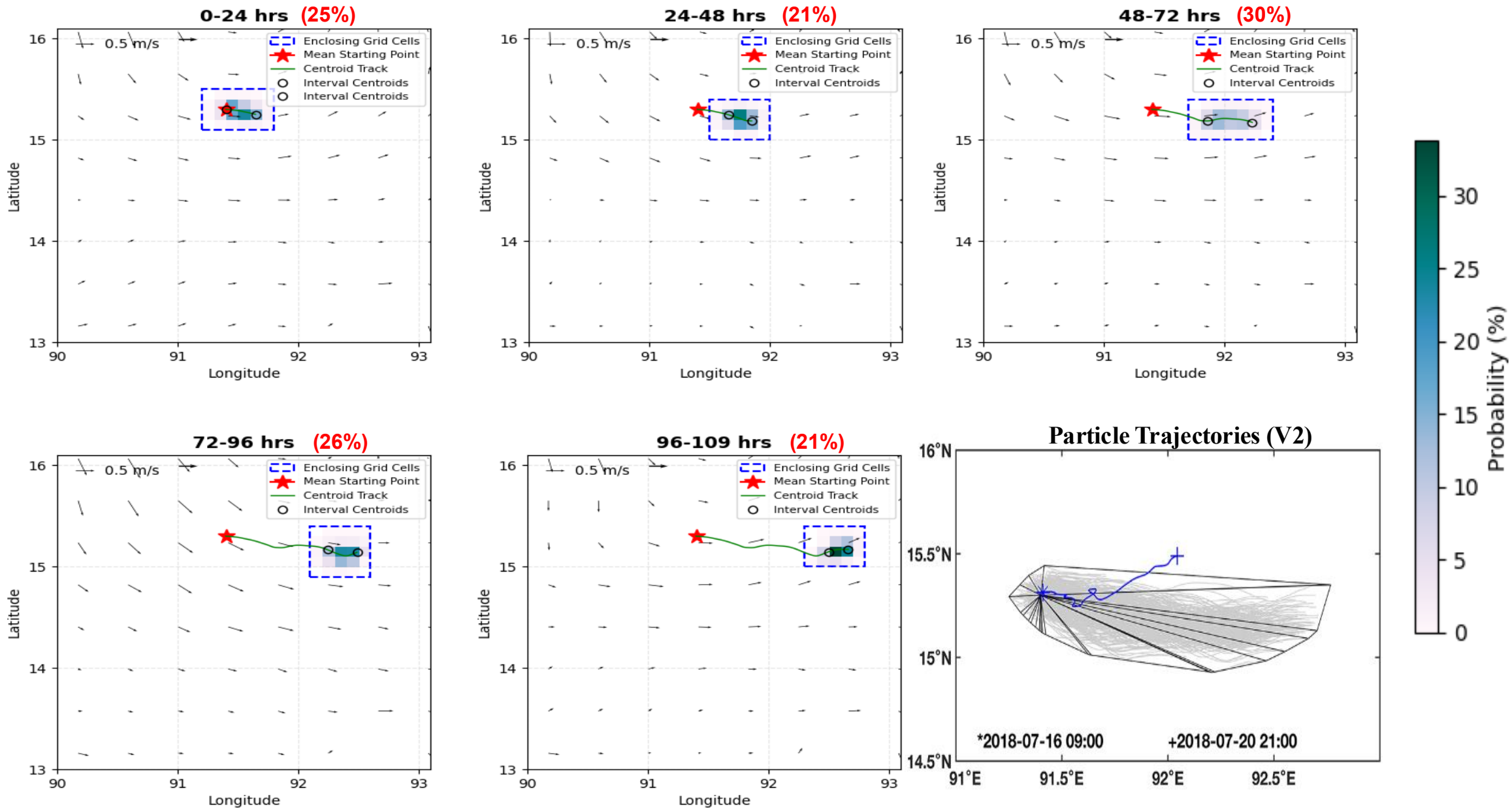
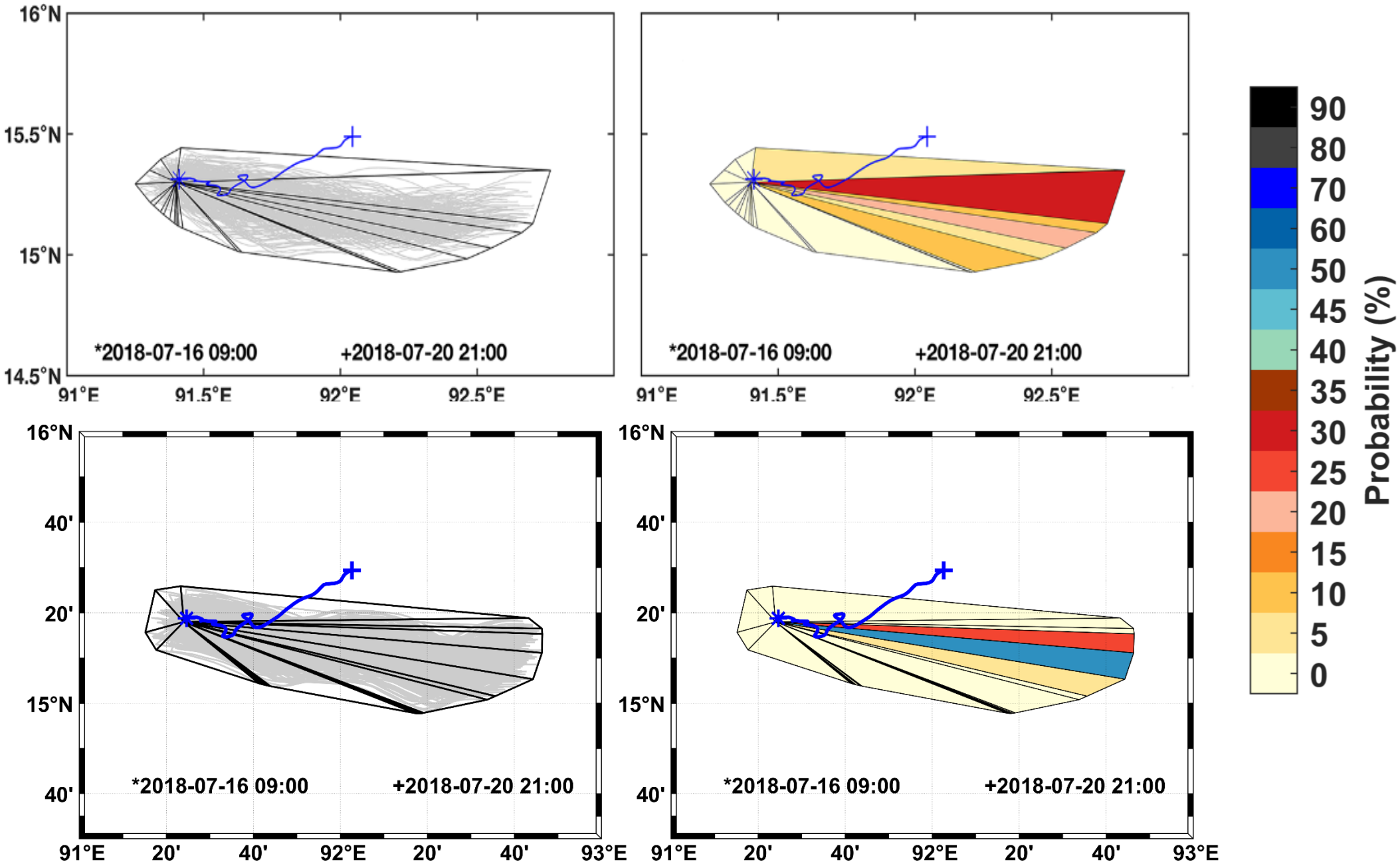


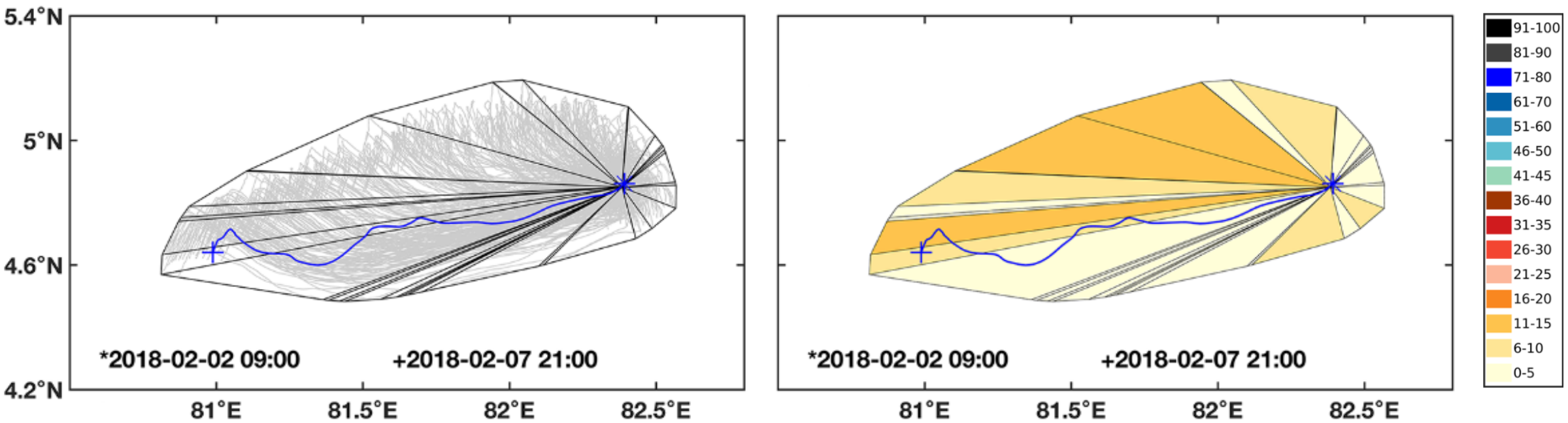
Simulated particle trajectories and probability calculation in SARAT versions 2 and 3: Case 1



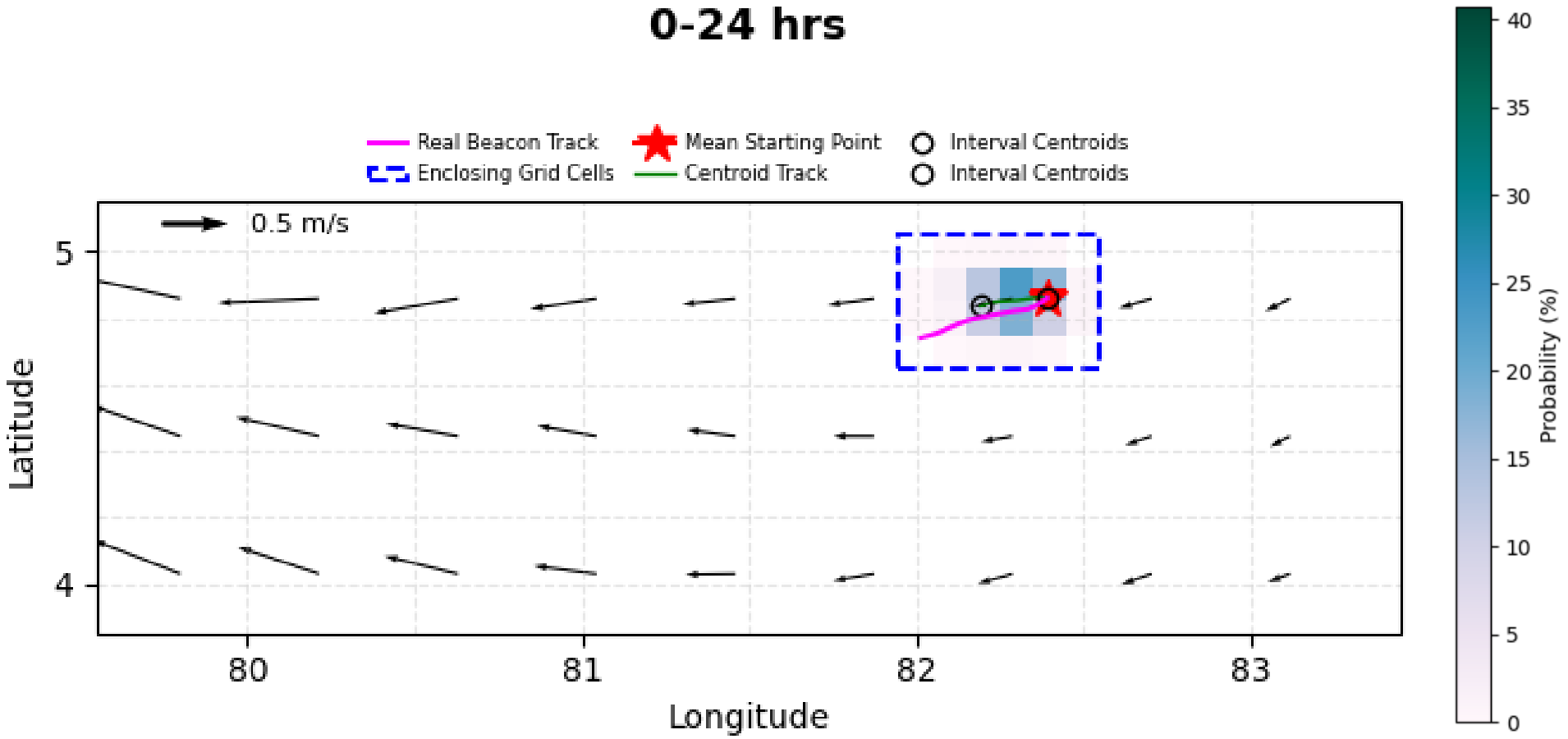
Comparison SARATV2 and SARATV3 (pov @SARATV2) [Case1]



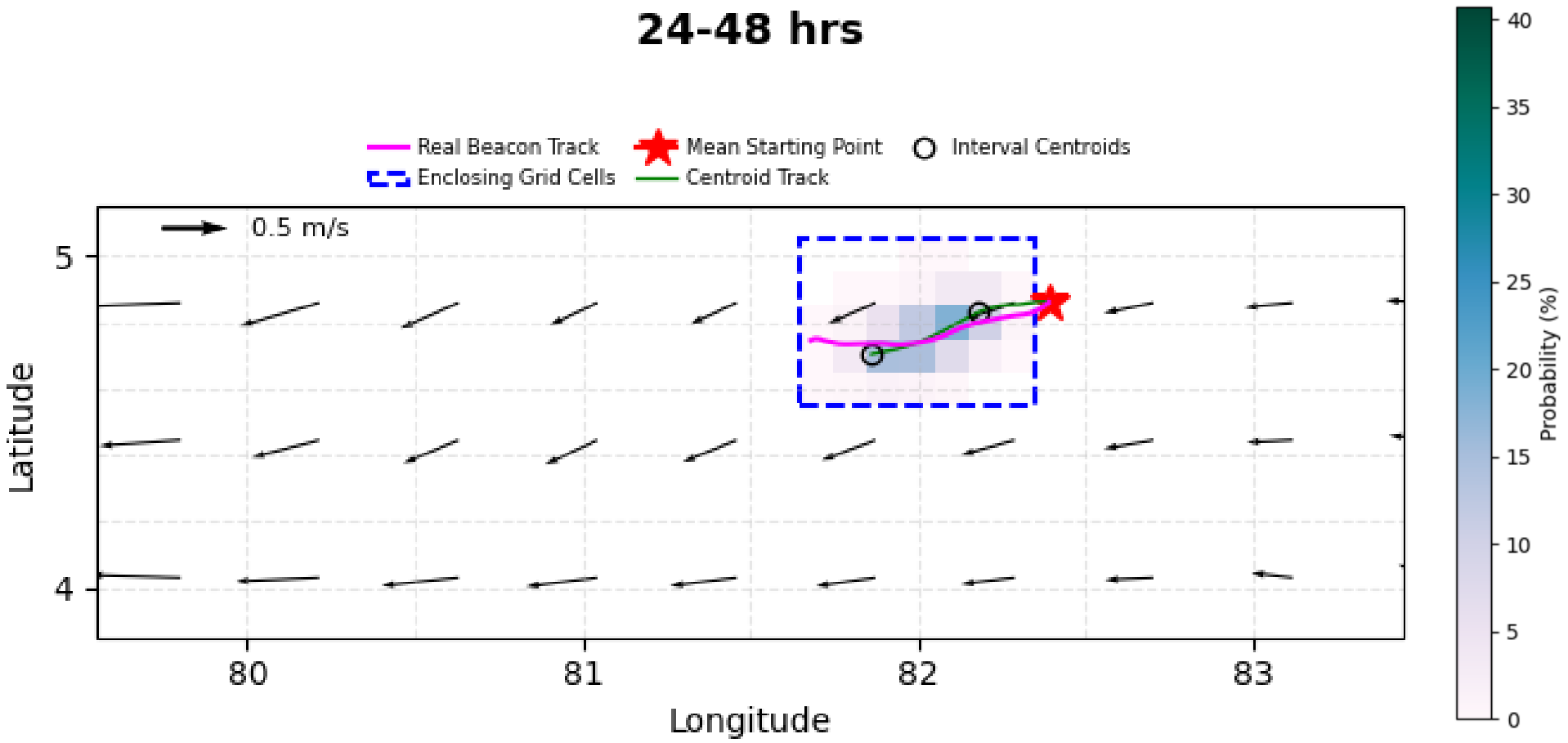
Simulated particle trajectories and probability calculation in SARAT version 2: Case 2



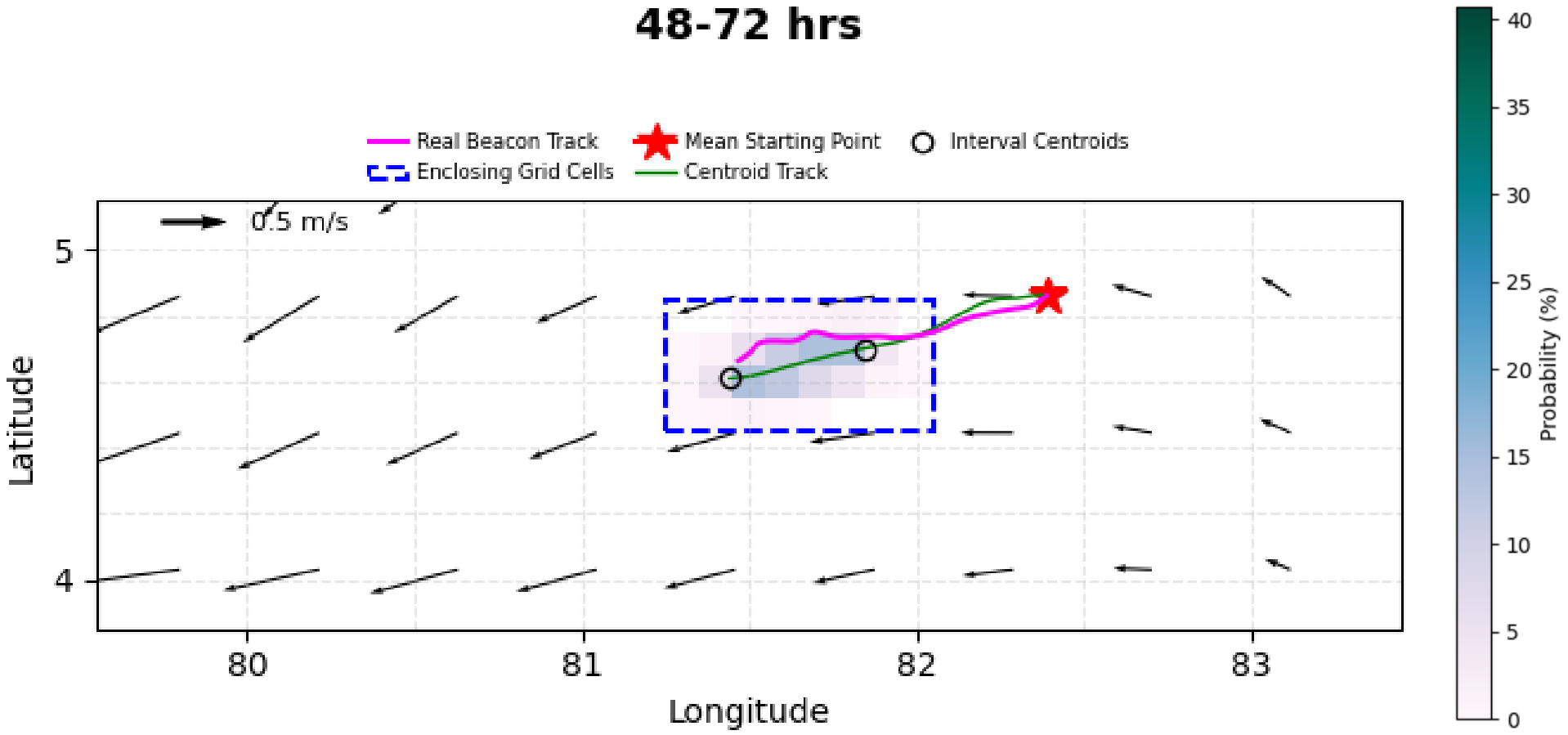
Time evolution of probability map in SARAT version 3: Case 2



Time evolution of probability map in SARAT version 3: Case 2



Time evolution of probability map in SARAT version 3: Case 2



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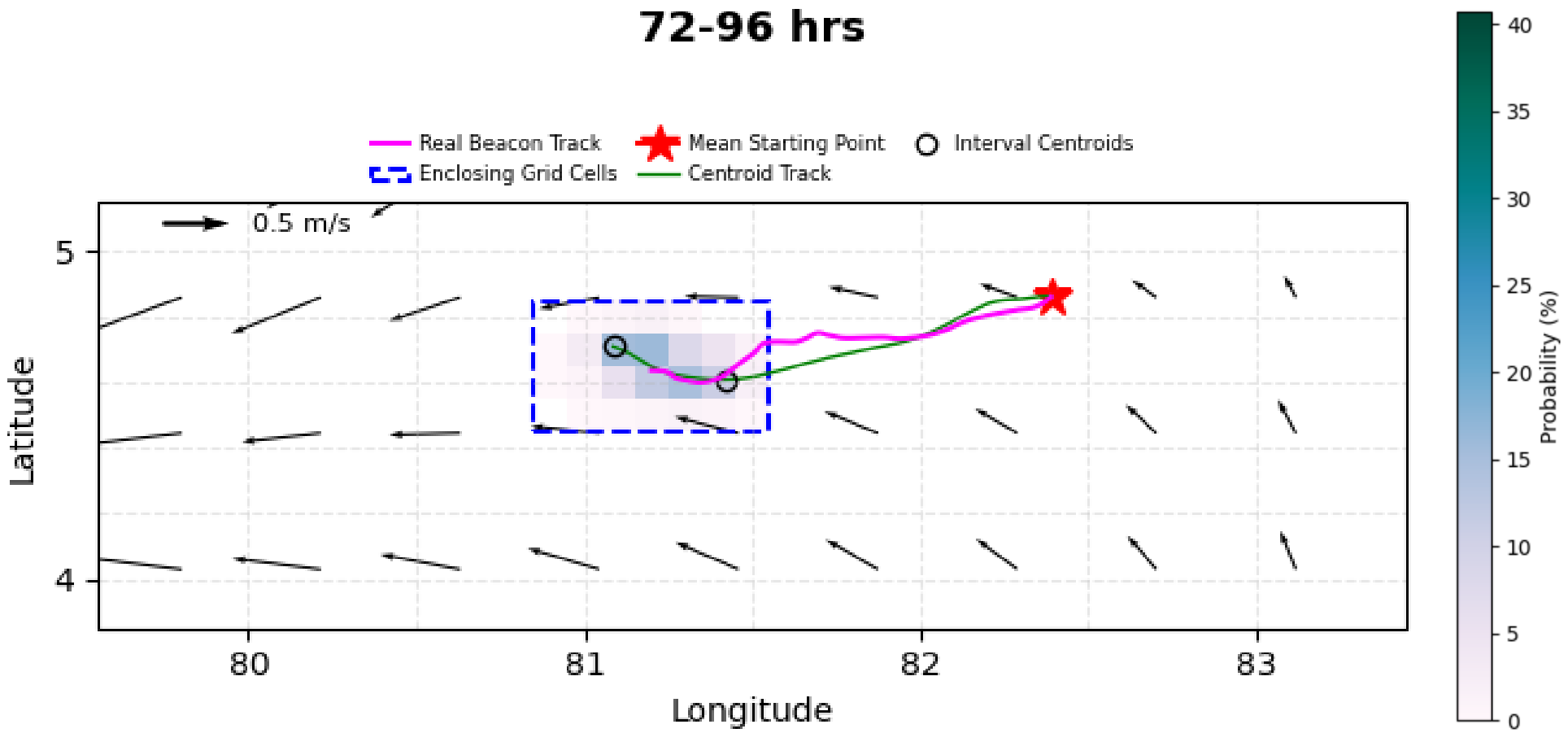

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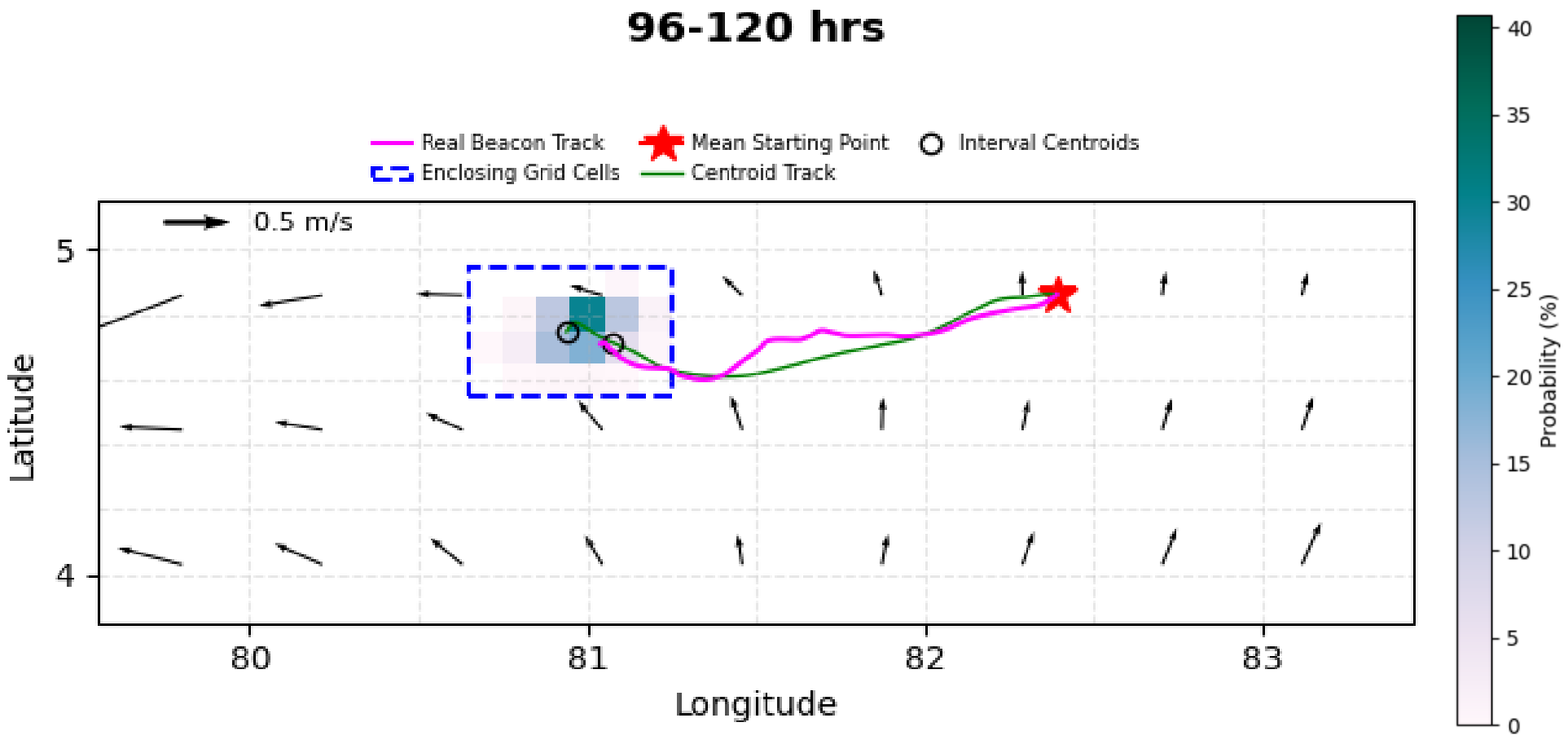

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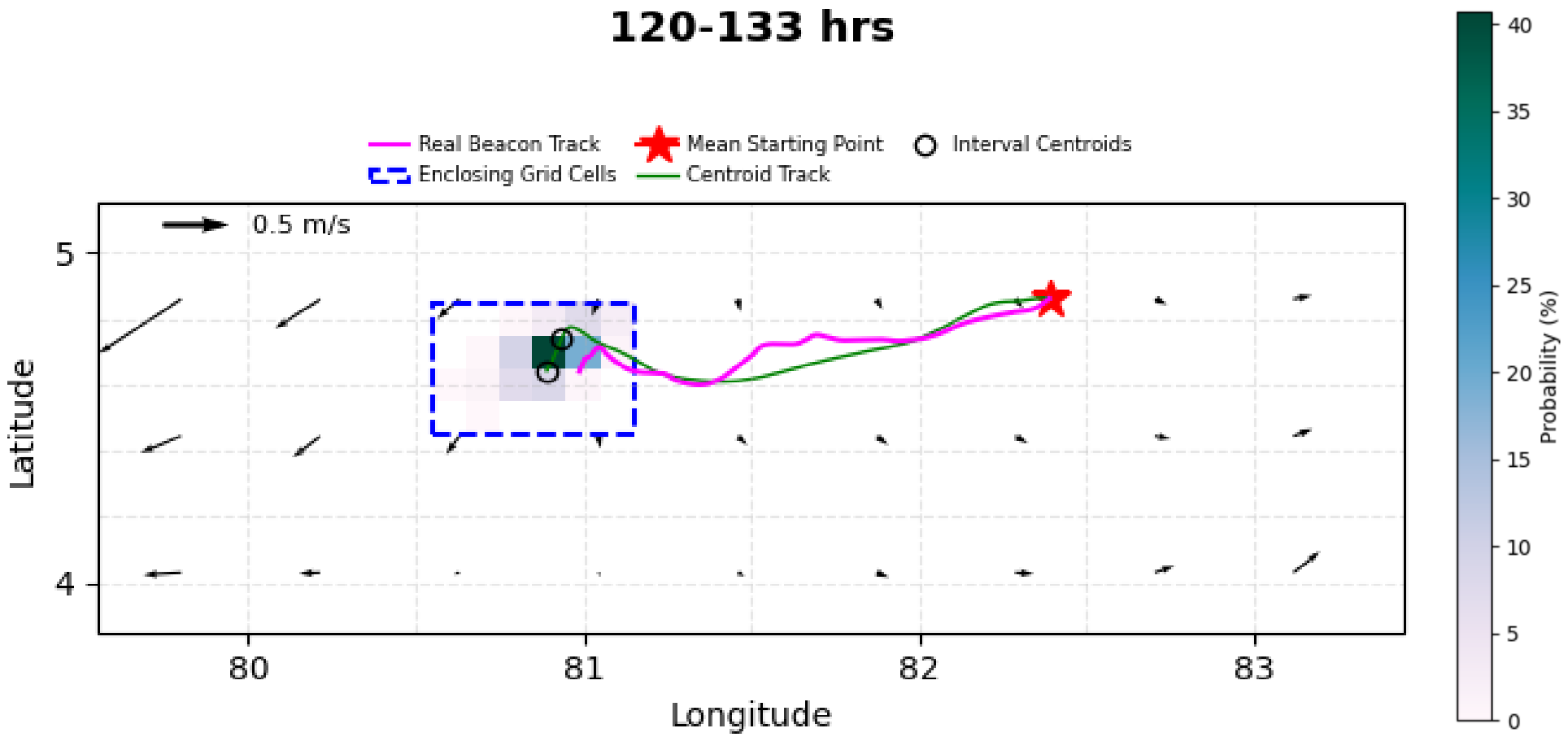
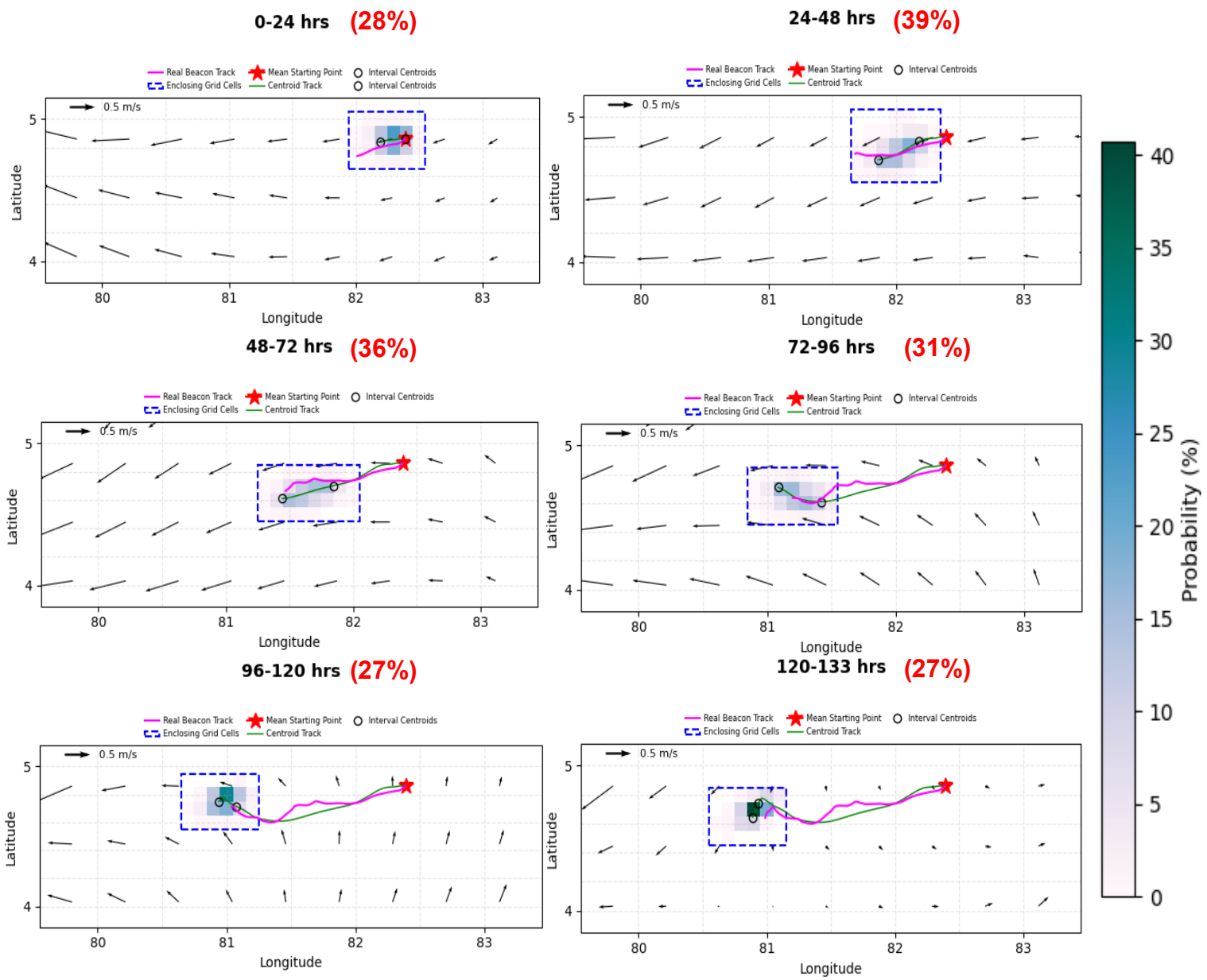
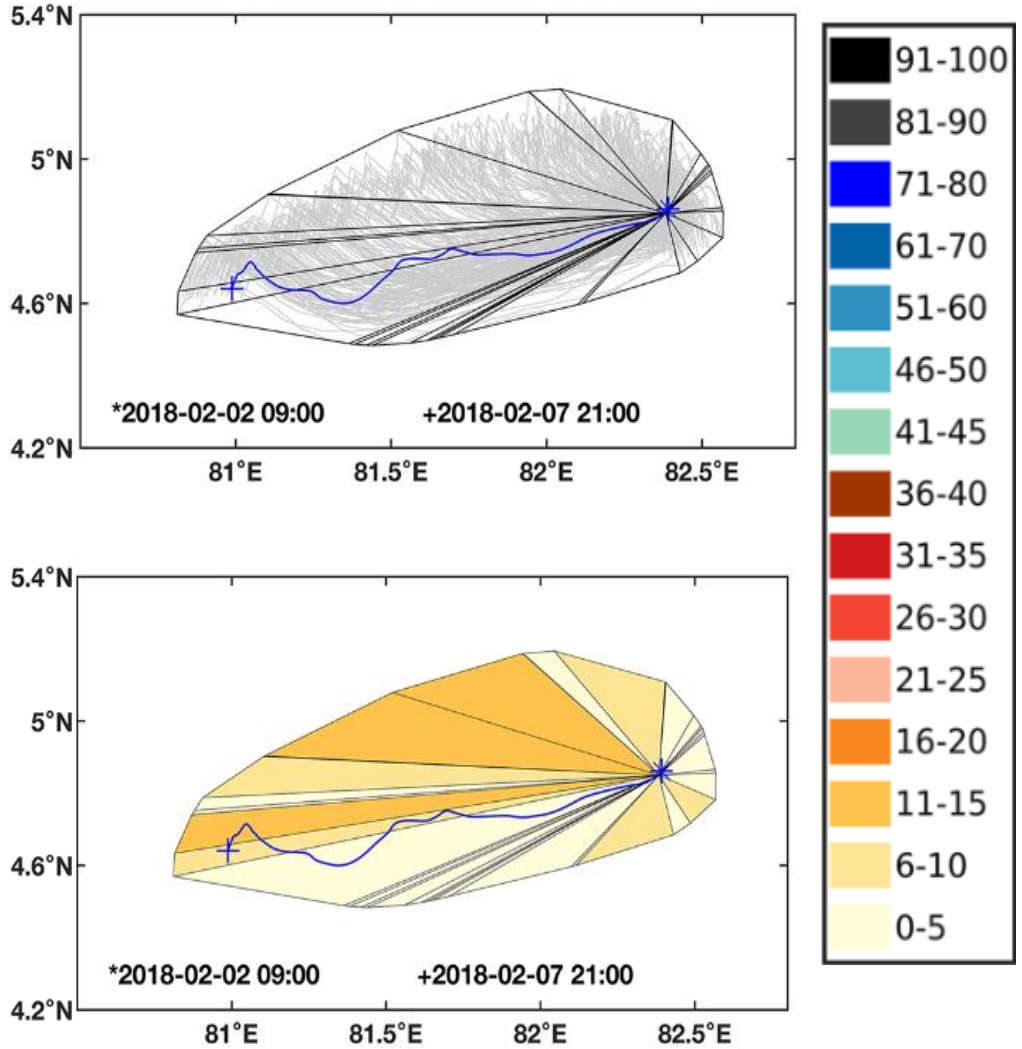
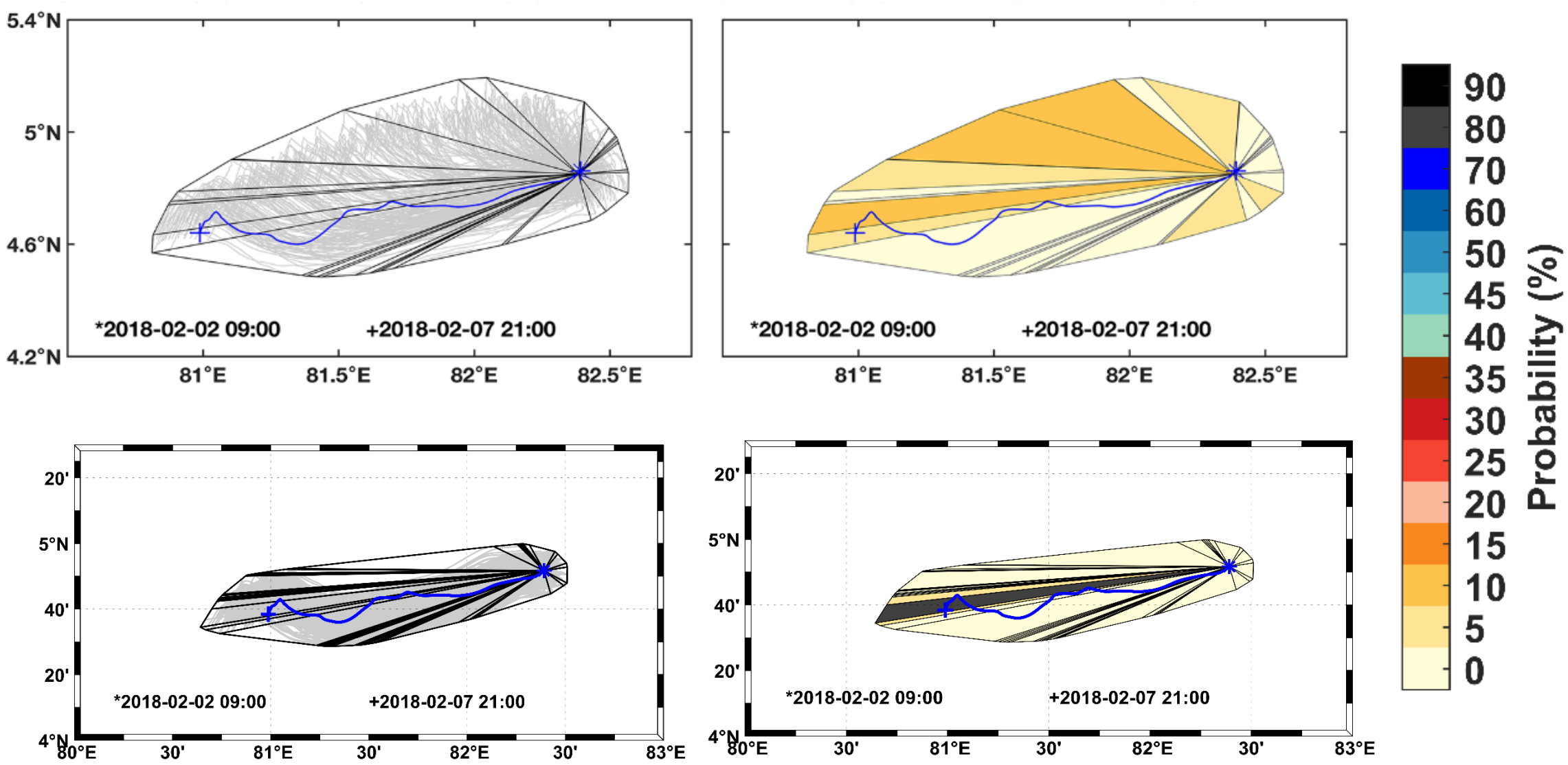


Simulated particle trajectories and probability calculation in SARAT versions 2 and 3: Case 2

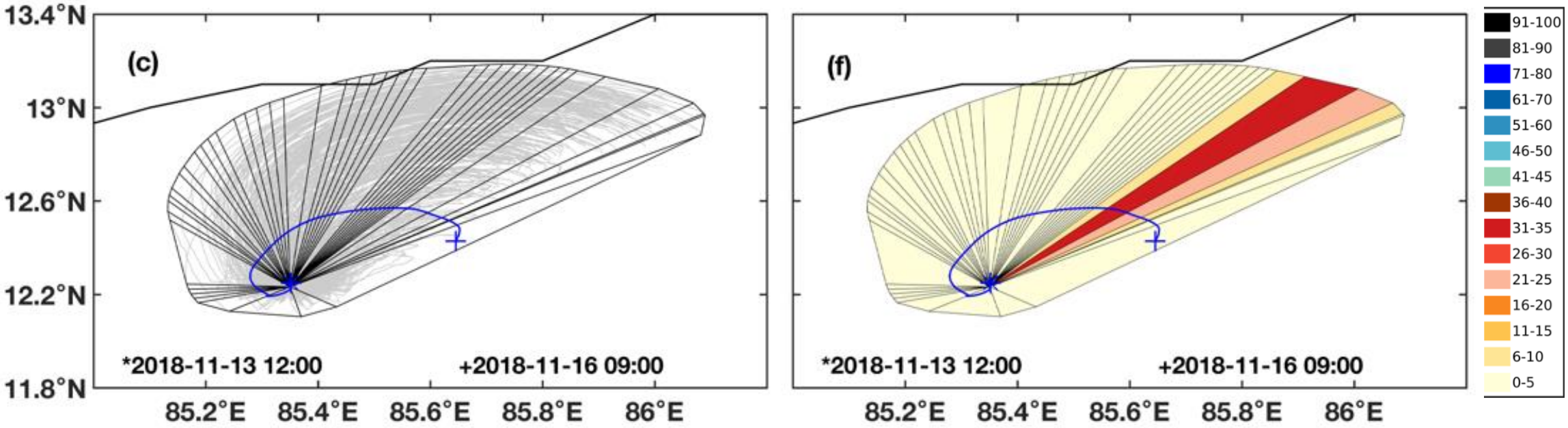




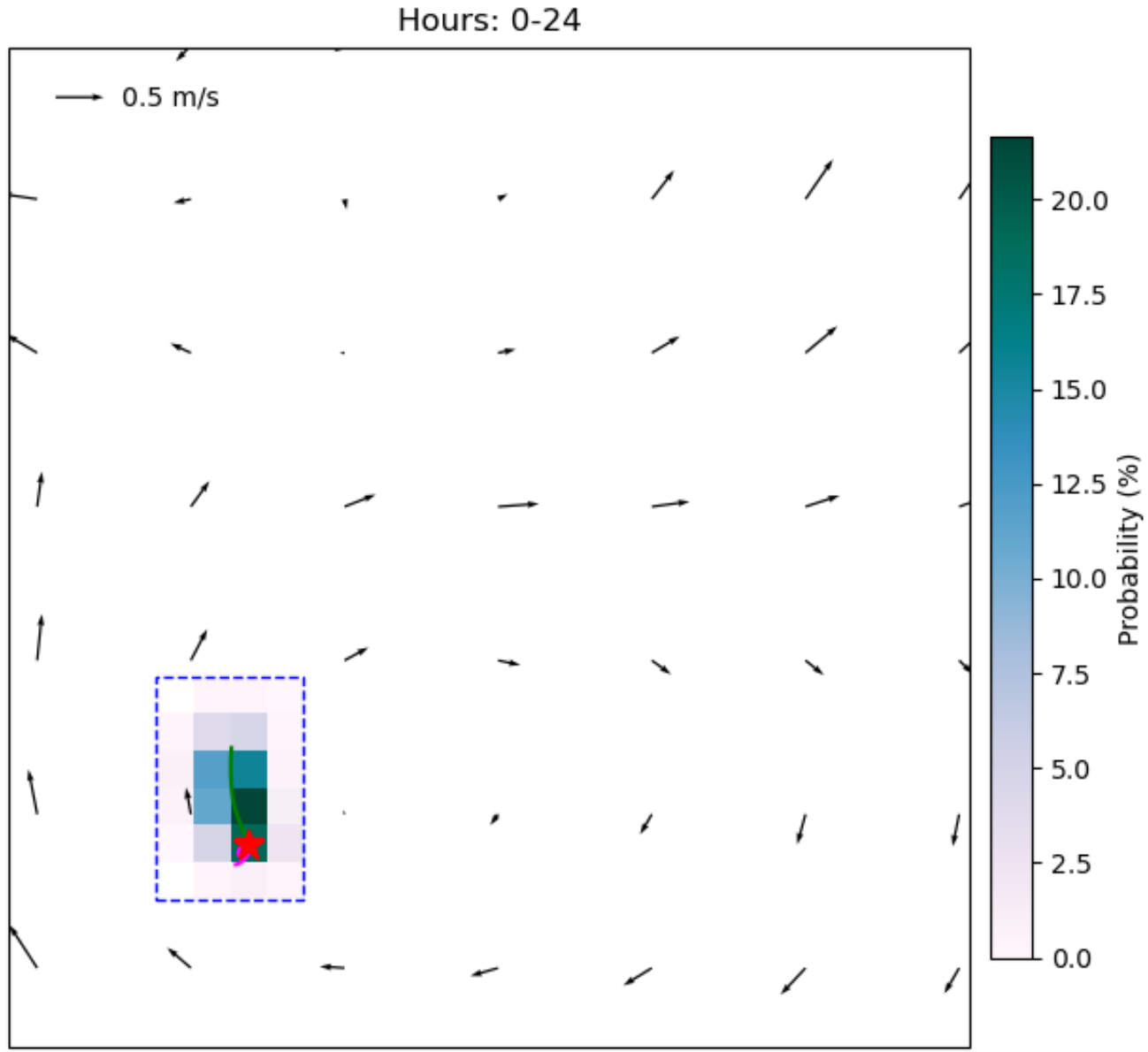
Comparison SARATV2 and SARATV3 (pov @SARATV2) [Case2]



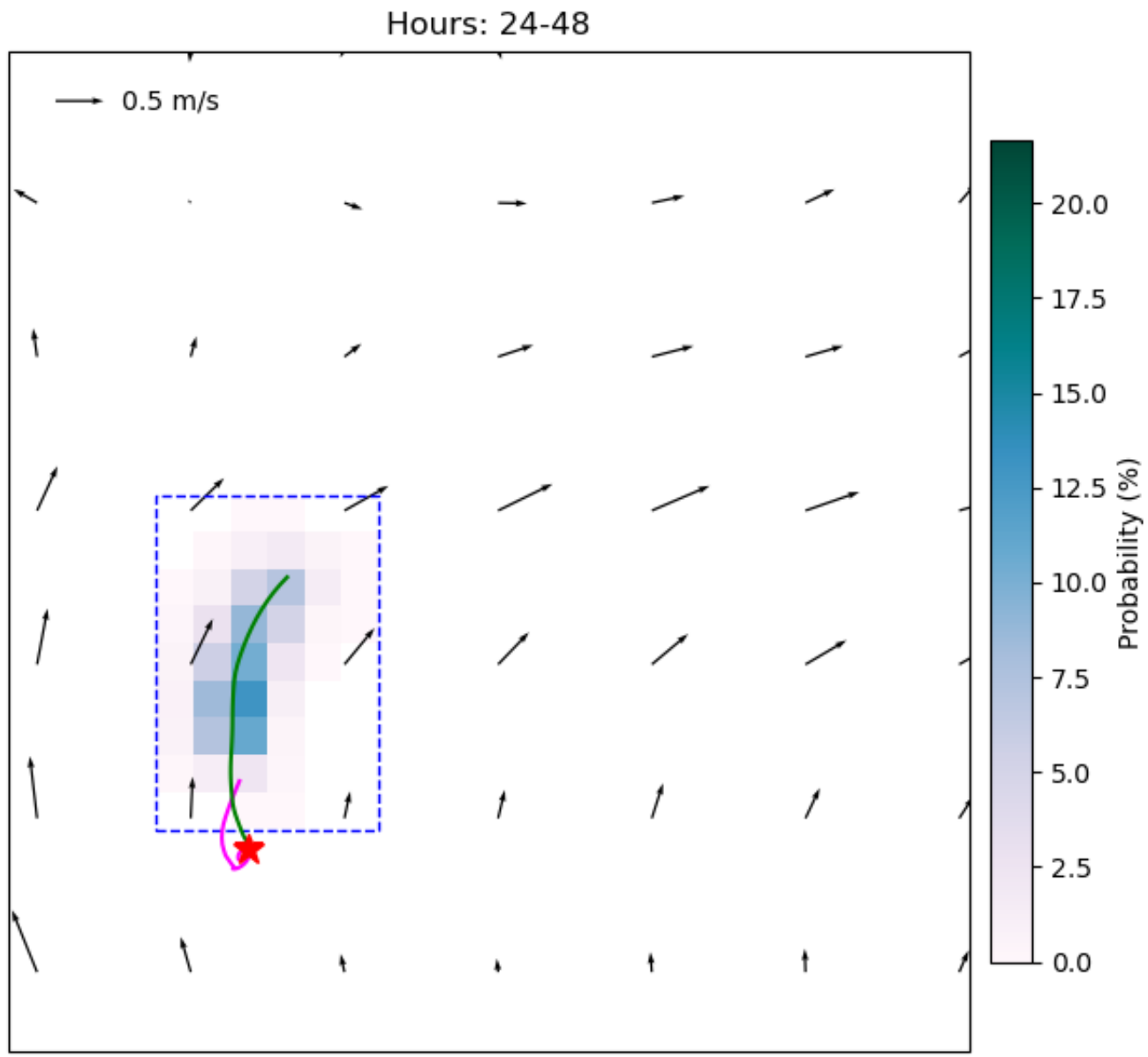
Simulated particle trajectories and probability calculation in SARAT version 2: Case 3



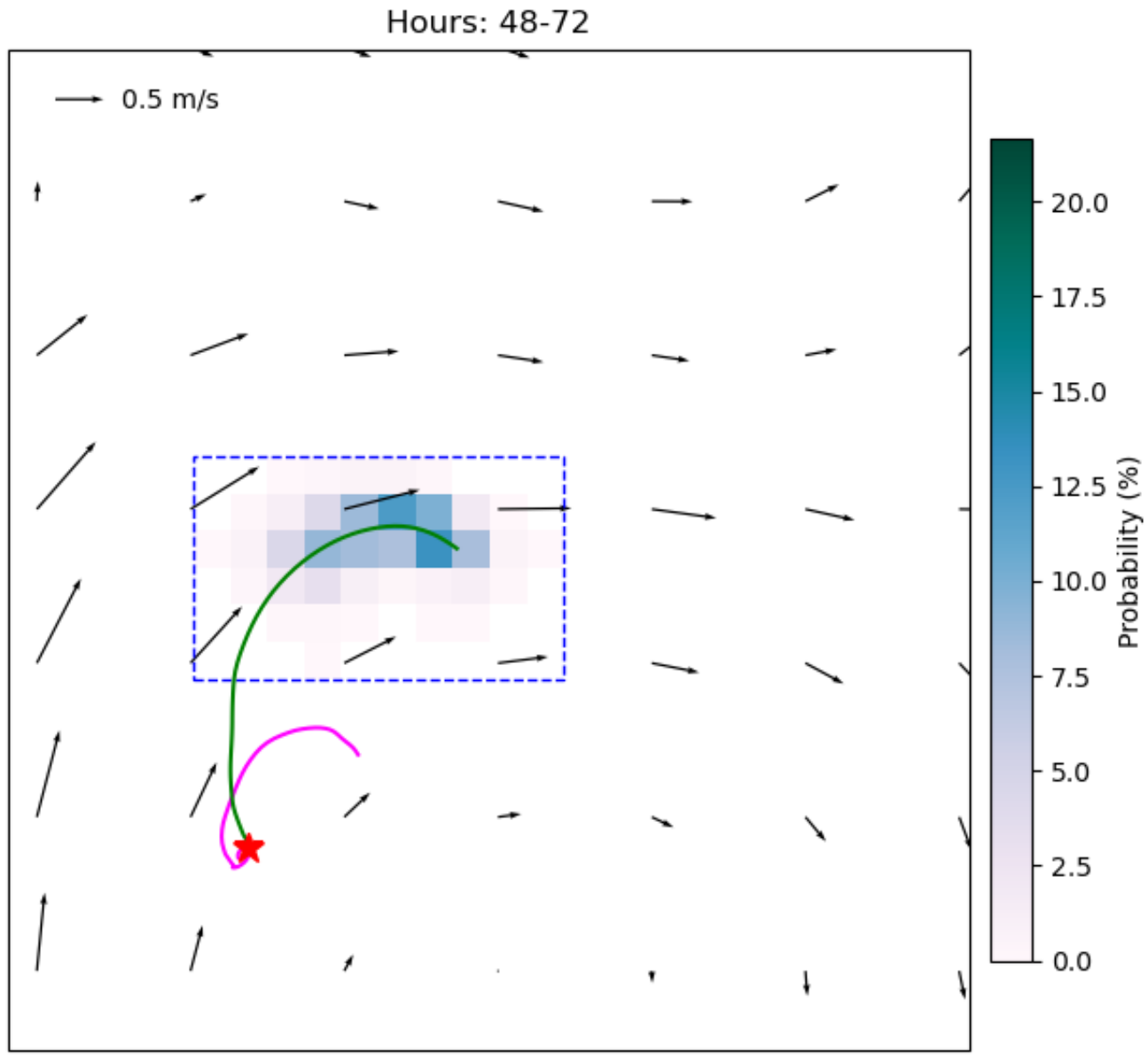
Time evolution of probability map in SARAT version 3: Case 3



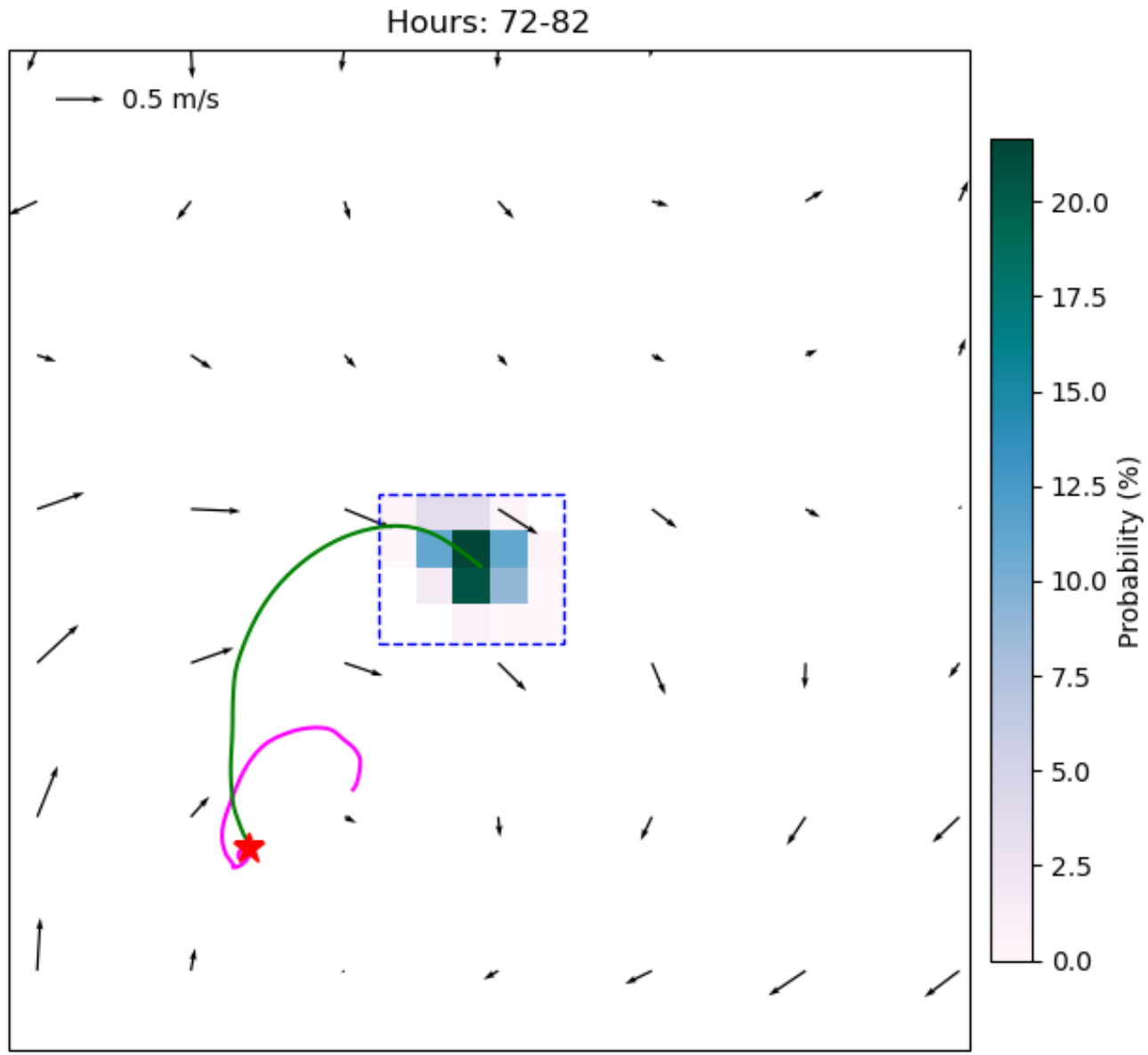
Time evolution of probability map in SARAT version 3: Case 3



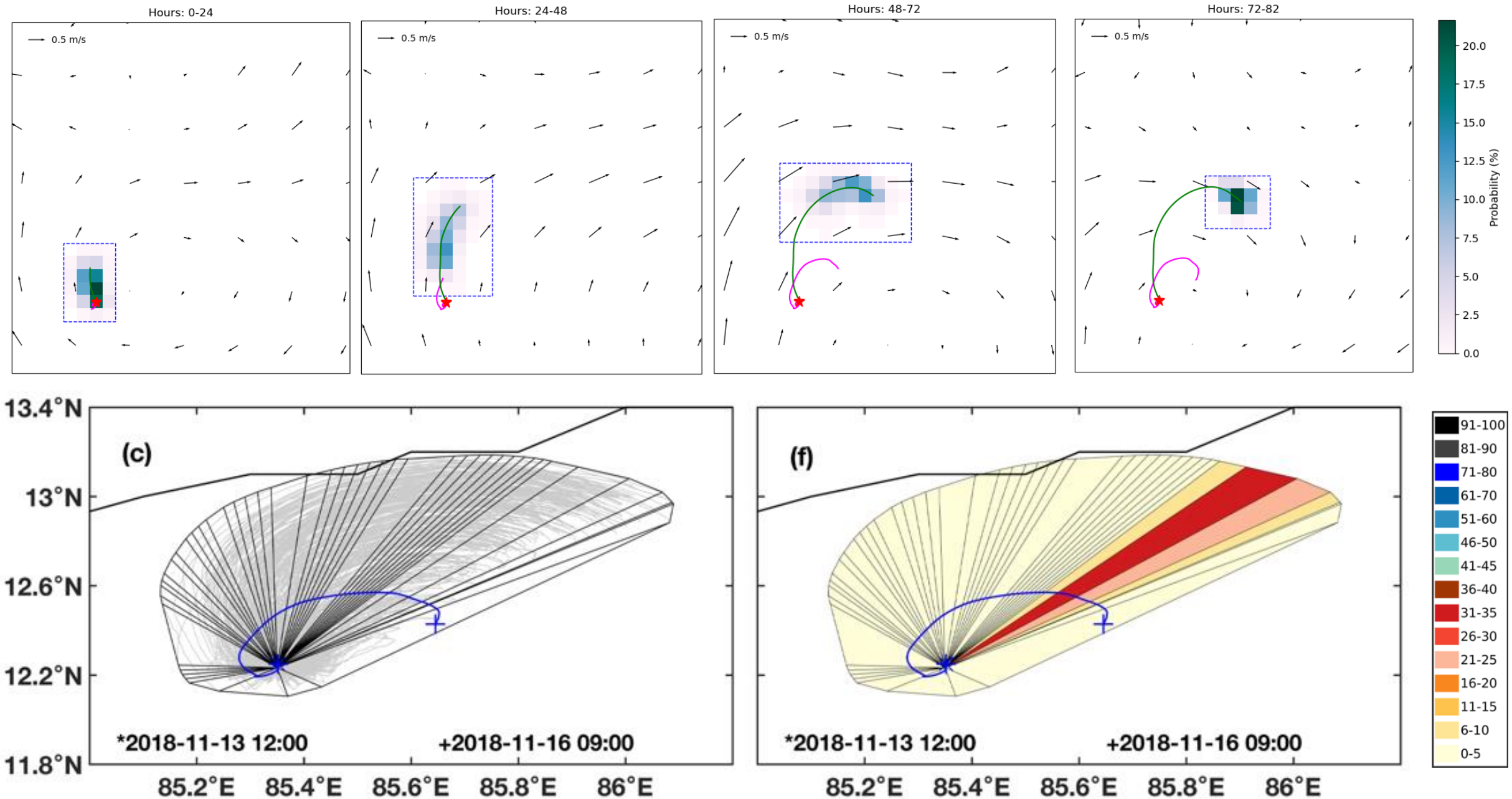
Time evolution of probability map in SARAT version 3: Case 3



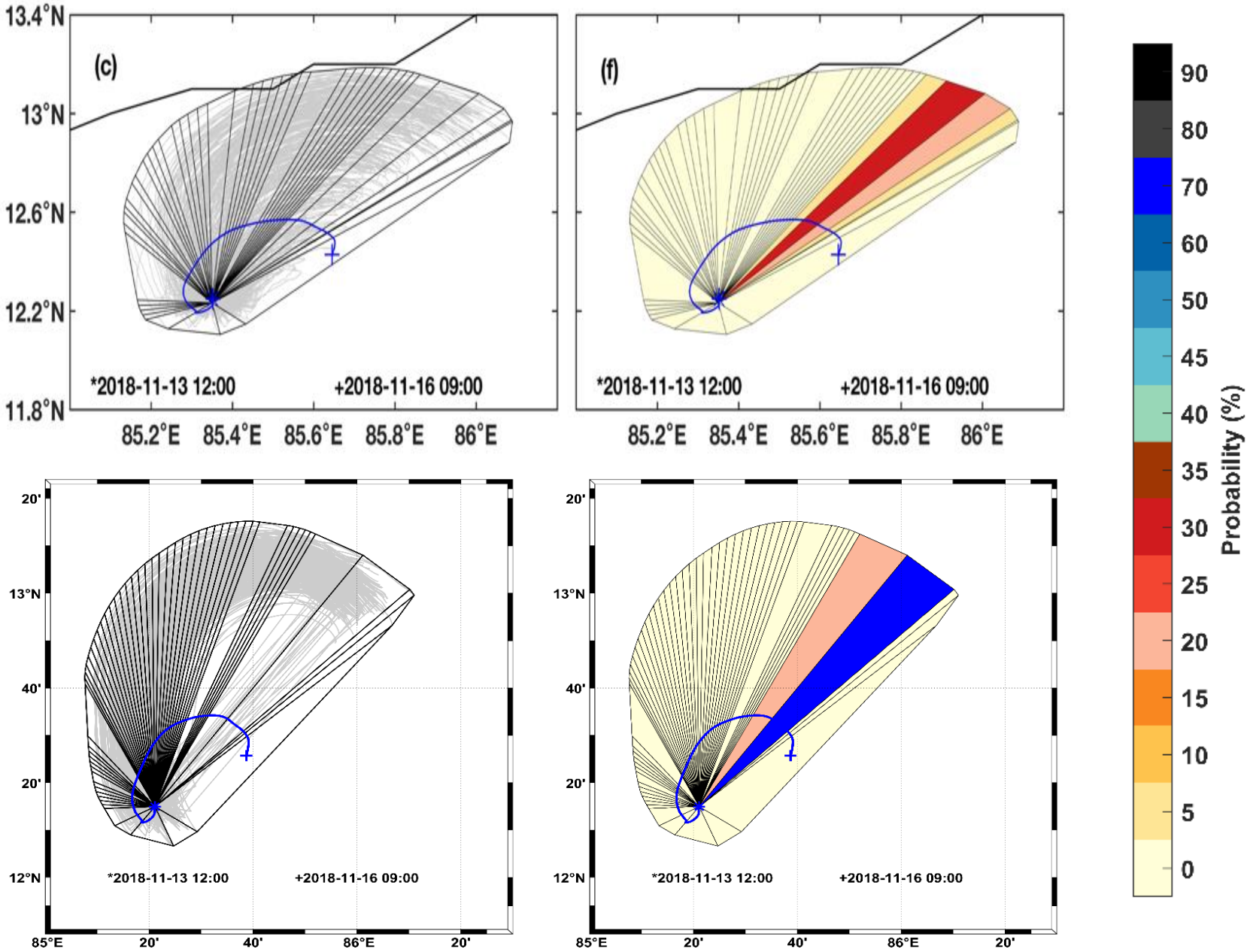
Time evolution of probability map in SARAT version 3: Case 3



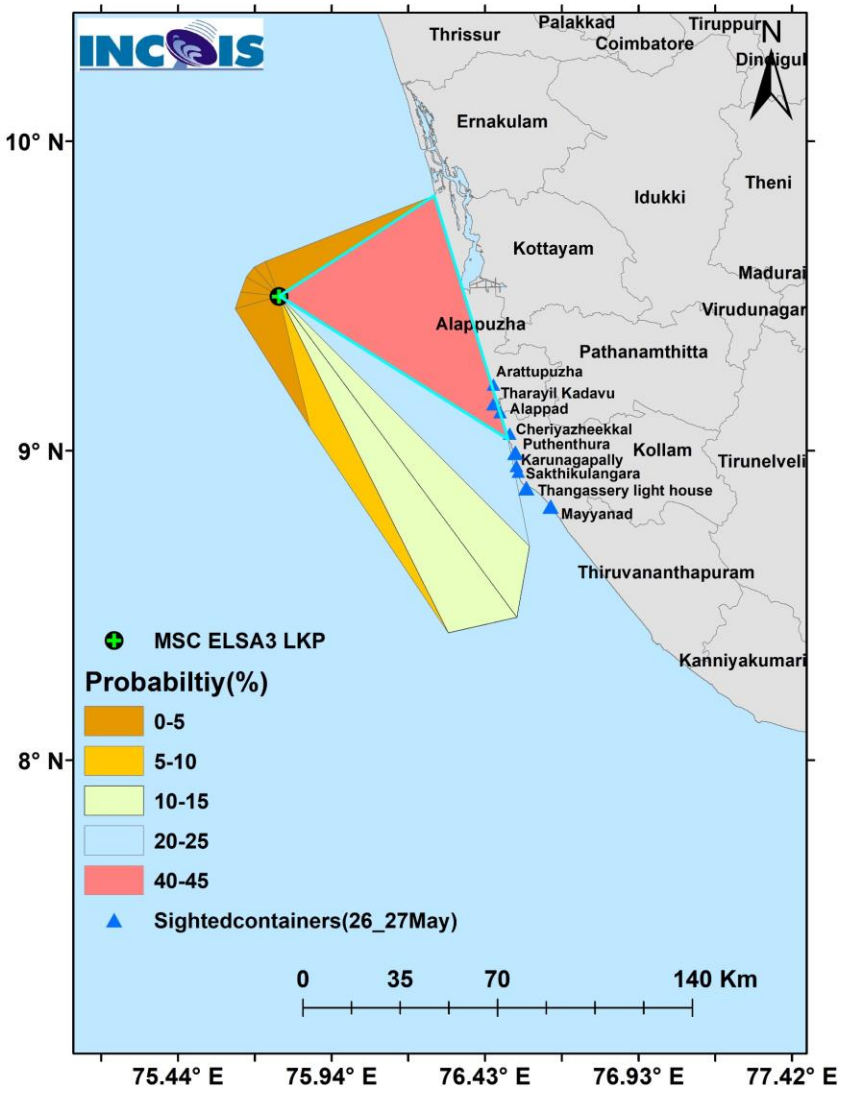
Simulated particle trajectories and probability calculation in SARAT versions 2 and 3: Case 3

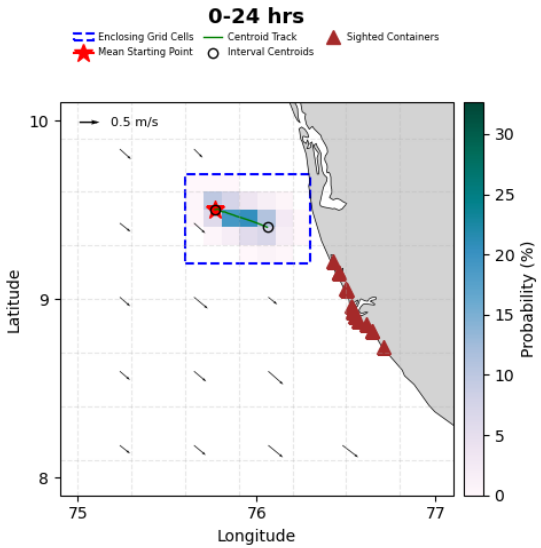


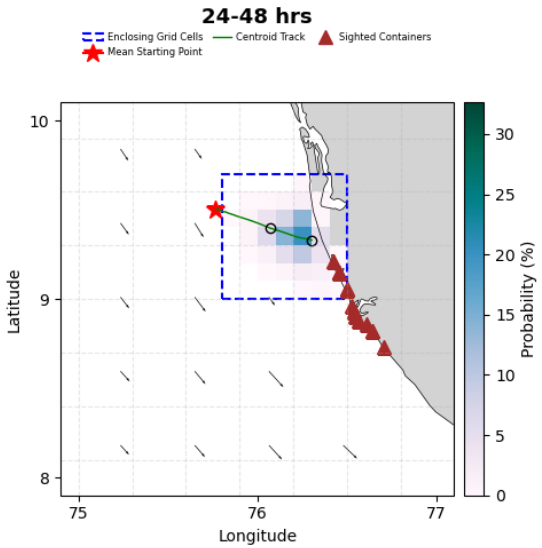
Comparison SARATV2 and SARATV3 (pov @SARATV2) [Case3]

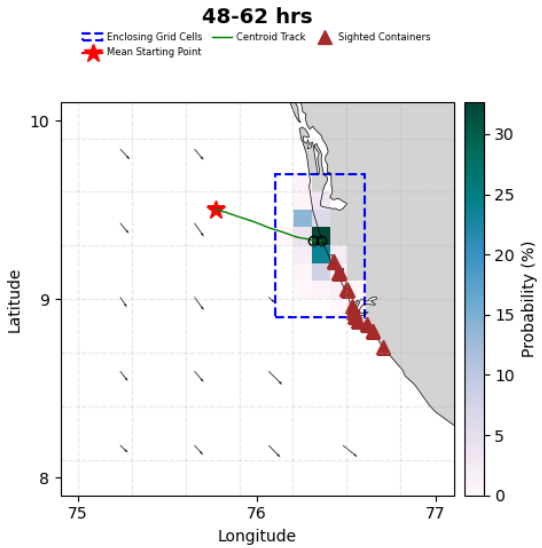


Simulated particle trajectories and probability calculation in SARAT versions 2 and 3: Case 3 (Containers drift case of May 2025)

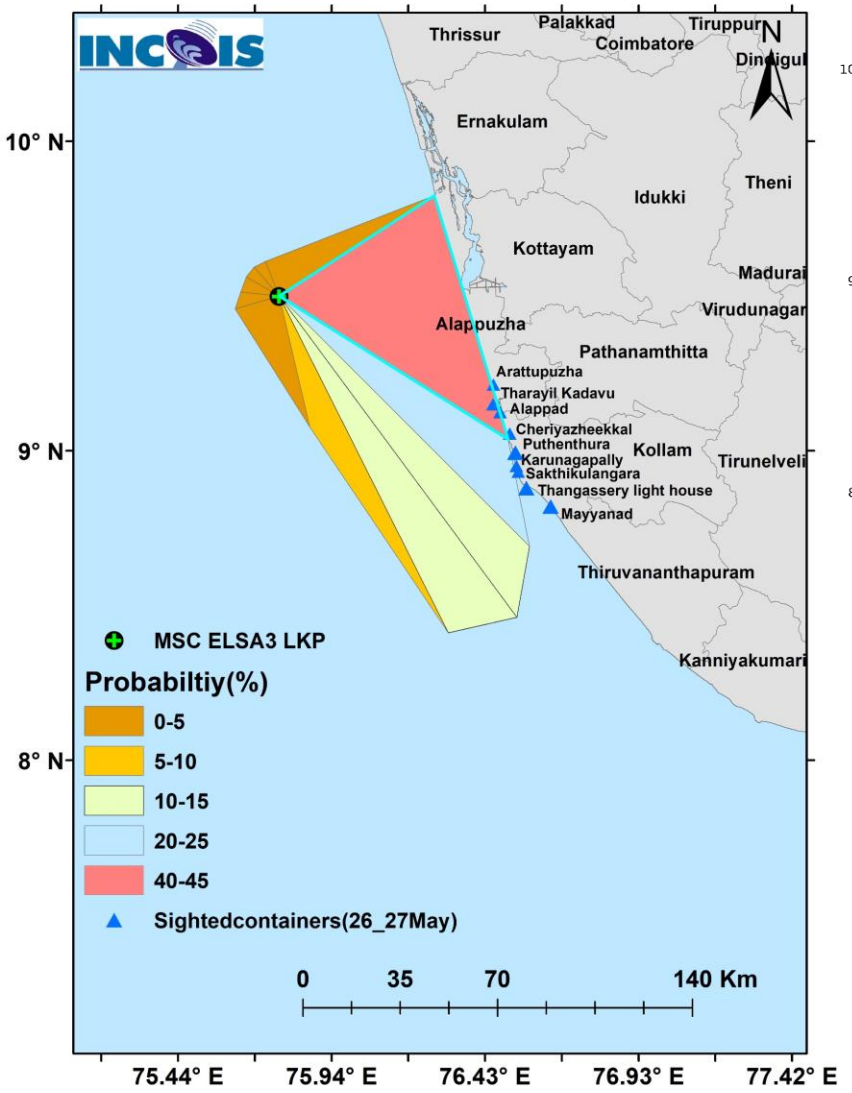


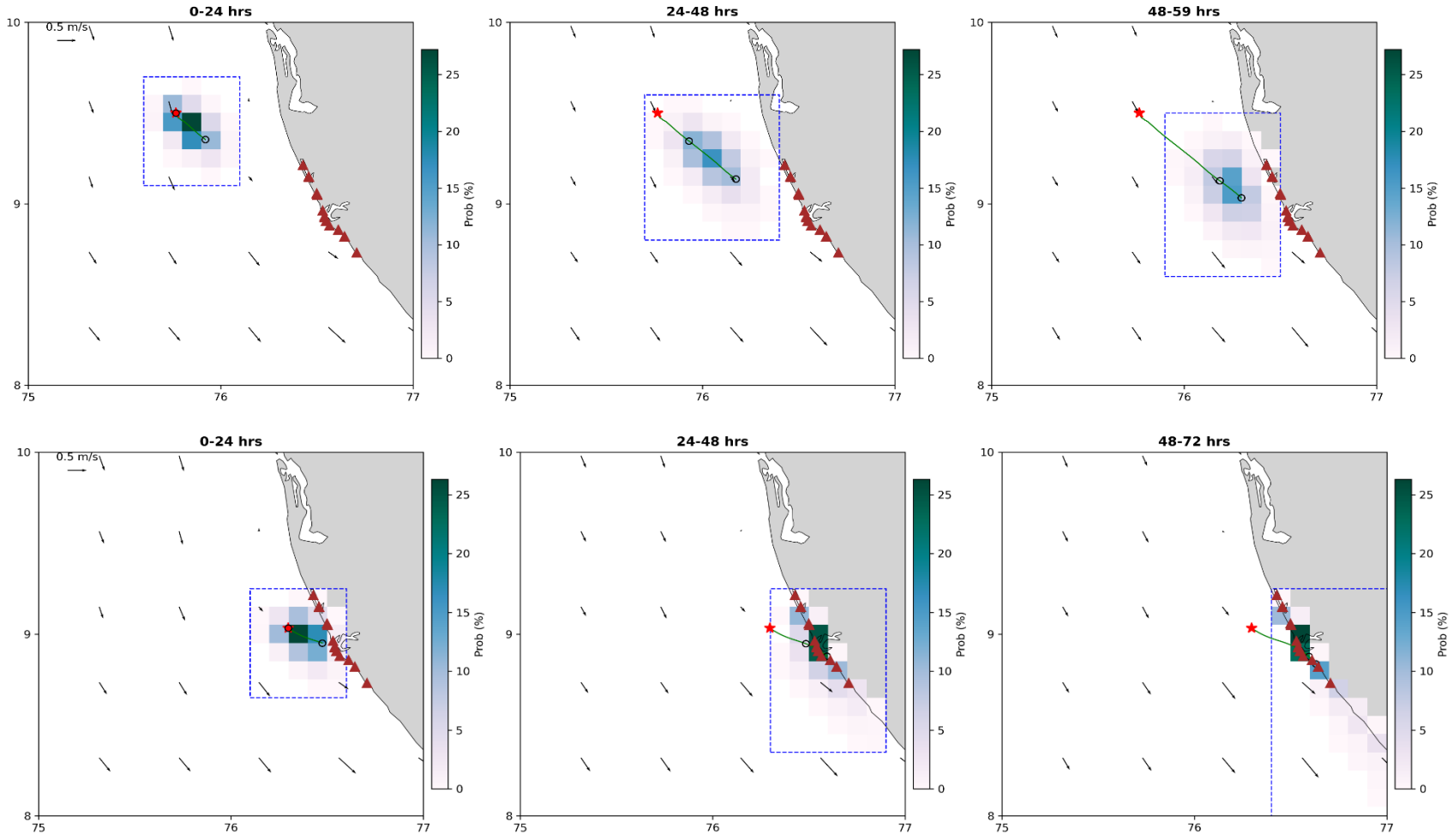






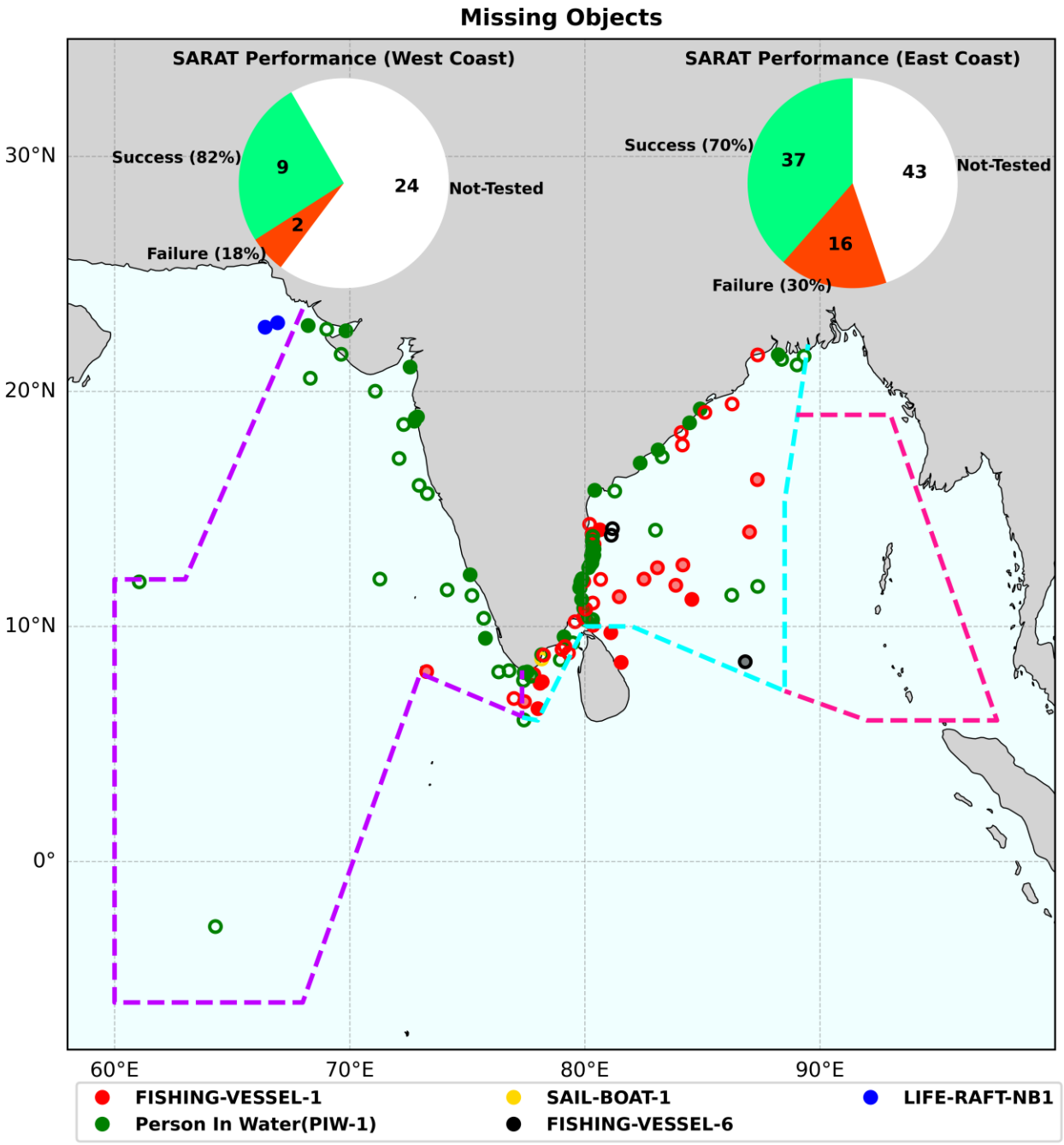
Simulated particle trajectories and probability calculation in SARAT versions 2 and 3: Case 3 (Containers drift case of May 2025)





SARAT validation using real use cases from the ICG for the period Oct 2024 – Oct 2025

- SARAT usage is more on the east coast. Also, the success rate.
- Most of the incidents on the east coast involve humans and fishing vessel
- On the west coast, most of the incidents involve humans. The percentage of untraced objects is very high.
- Overall, most of accidents occur in the coastal areas

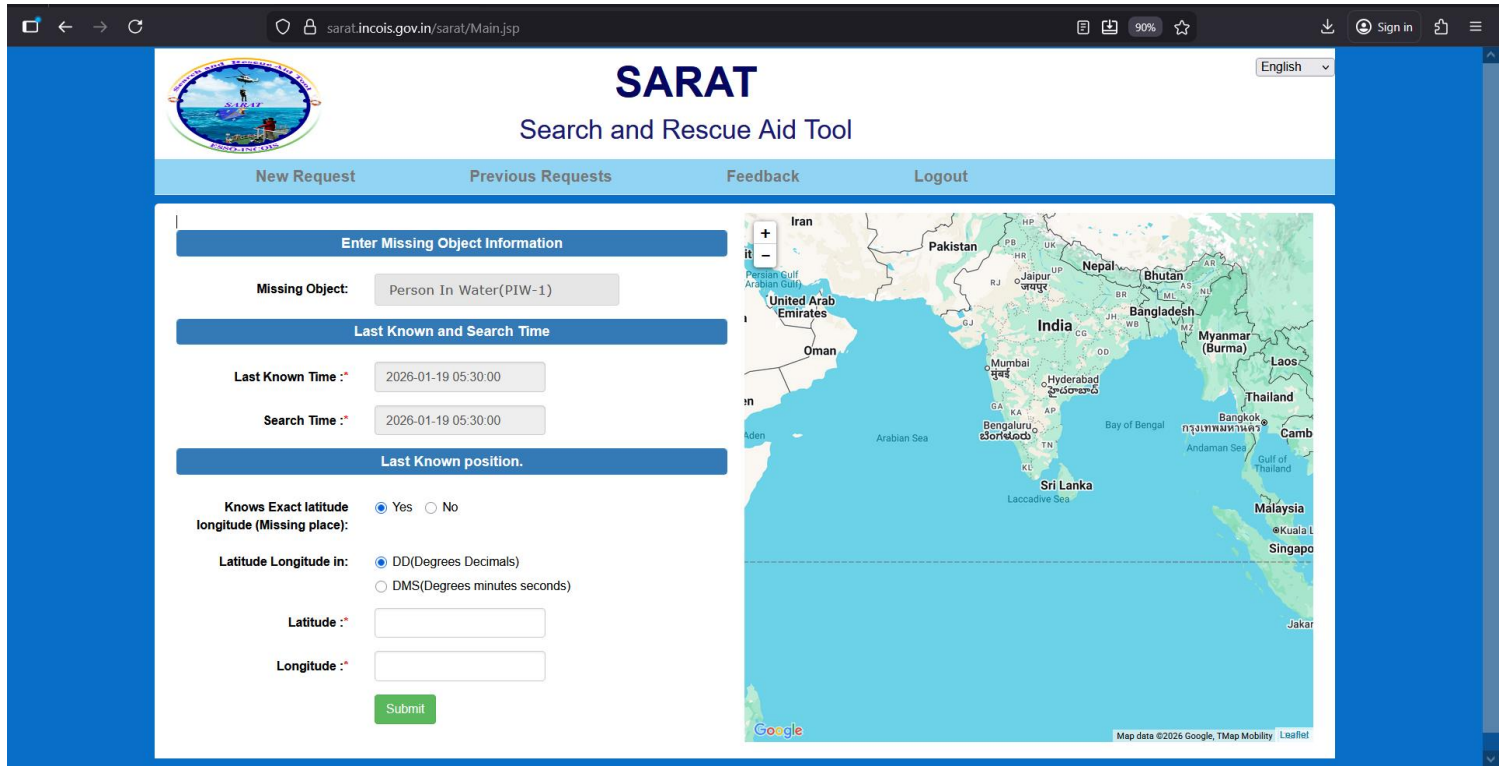


Major model improvements SARAT version 3

- Sprinkling of the particles can now be controlled
- Probabilities are calculated for the selected regular intervals
- Shows the background currents averaged for the respective intervals
- Adds the time evolution of probabilities in uniform grid cells (0.1*0.1)
- Provides the maximum search area bounding box for the interval
- Respective centroids of the interval and mean trajectory provide the evolution of search area.
- Search area is drastically reduced decreasing the search time.
- Aids in a better planning of search operations
- Addition of a new object category called ‘zero-leeway’ for small objects such as buoys. The wind effect on such objects is supposed to be minimal
- Stranding of particles

Proposed visualization improvements in SARAT version 3

- Inclusion of Degree Decimal units for the LKP
- Simulation duration instead of the Search Time
- Addition of seeding duration
- Design of the user interface improvements to adapt the new model improvements.
- Integration of the coastal map in the outputs (web and pdf)
- Inclusion of the LKP1 and LKP2 for objects missing from a moving ship or so.
- Addition of FAQ page on the web application



Steps for the operational implementation of SARAT version 3

- Restriction of the seeding duration to 12 hours if the Simulation period is more than a day. Otherwise, 3 hours or 1 hour minimum. The seeding time should be optimally given so that it is long enough but at the same time takes the uncertainty and time-locking in the forecasted currents into account
- Modified and re-compiled code needs to be used for the above
- Overhauling of the presentation of probabilities and their evolution with time
- In addition of version 3 outputs, earlier version 2 output will also be provided both on the web application and the pdf for a comparison and smooth transition
- Segregation of SARAT environments carefully to ensure current version is still accessible to users till the new version is made online.
- Design of appropriate user interface improvements in the web application to adapt the new probability calculations.
- Modifications to the web application codes to reflect the proposed visualization changes

Steps for the operational implementation of SARAT version 3

- Translate the version 3 model improvements into the web application and conduct several test simulations to ensure the web application is doing what it is supposed to do
- Obtain the feedback of the ICG on the proposed version 3 and implement their suggestions. An online meeting needs to be conducted with the three MRCCs
- Adopt the best practices from other operational centres. NOAA and OCEAN-SAR of CMCC, SARMAP of the RSP Company.
- Demonstration of the implemented version 3 before the operationalization
- Preparation of a technical report detailing the improvements and an SOP document for the operational team
- Training the operational team and operationalization of SARAT version 3 after successful trials
- Continuation of version 2 parallel to the version 3 for a few months for a smooth transition
- Training workshop for the ICG personnel on SARAT version 3
- Revise SARAT research article and submit it to an international journal

Operational implementation of SARAT version 3: Resource requirements and other arrangements

- A dedicated computer scientist for the operationalization period
- Server (Existing SARAT server could also be utilized).
- Formalization of provision of regular feedback (quarterly or so) from the ICG

THANK
YOU